

4

MEASUREMENT AND SPACE

TRIGONOMETRY

In the second century BCE, the Greek astronomer Hipparchus could calculate distances to the moon and the Sun and he was the first scientist to chart the positions of over 850 stars. How was he able to achieve this over 2100 years ago? Hipparchus started a branch of mathematics called **trigonometry**, meaning 'triangle measure', which uses angles, triangles and circles to calculate lengths and distances that cannot be physically measured. Trigonometry is used widely today in engineering, surveying, navigation, astronomy, electronics and construction.



Chapter outline

	Working mathematically				
4.01 The sides of a right-angled triangle	U	F			C
4.02 The trigonometric ratios	U	F		R	C
4.03 Similar right-angled triangles	U	F		R	C
4.04 Trigonometry on a calculator	U	F			
4.05 Finding an unknown side	U	F	PS		
4.06 Finding more unknown sides	U	F	PS		
4.07 Finding an unknown angle	U	F	PS		
4.08 Angles of elevation and depression	U	F	PS		
4.09 Bearings	U	F		R	C
4.10 Problems involving bearings	U	F	PS		C

Wordbank

adjacent side In a right-angled triangle, the side that is next to a given angle and pointing to the right angle

angle of depression The angle of looking **down**, measured from the horizontal

angle of elevation The angle of looking **up**, measured from the horizontal

bearing The angle used to show the direction of one location from a given point

minute (') A unit for measuring angle size, $\frac{1}{60}$ of a degree

opposite side In a right-angled triangle, the side that is facing a given angle and not one of its arms

theta (θ) A letter of the Greek alphabet used as a variable for angles

trigonometric ratio The ratio of 2 sides in a right-angled triangle, for example, sine is the ratio of the opposite side to the hypotenuse

Quiz
Wordbank 4

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (16):

4.01 The sides of a right-angled triangle
4.02 The trigonometric ratios
4.04 Trigonometry on a calculator • Rounding angles
4.05 Finding an unknown side • Trigonometry
4.06 Finding the hypotenuse
4.07 Finding an unknown angle • Trigonometry
4.08 Angles of elevation and depression • Angle of depression 1 • Angle of depression 2
4.09 Three-figure bearings
4.10 Bearings • Trigonometry 2 • True bearings • Problems involving bearings

Twig videos (3):

SkillCheck Measuring the Earth • The tunnel of Samos • Distance to the Sun and Moon

Quizzes (5):

- Wordbank 4
- SkillCheck 4
- Mental skills 4
- Language of maths 4
- Test yourself 4

Worksheets (9):

4.03 Investigating the tangent ratio
4.04 Trigonometric calculations
4.06 Finding an unknown side
4.07 Finding an unknown angle • Trigonometry problems
4.09 A page of bearings • NSW map bearings • 16 points of the compass
Mind map Trigonometry (Advanced)

Puzzles (7):

4.02 Trigonometry match-up
4.04 Trigonometry squaresaw
4.06 Trigonometry equations 1
4.07 Trigonometry squaresaw • Trigonometry: Finding angles • Trigonometry equations 2
4.09 Bearings match-up
Language of maths Trigonometry crossword

Technology (1):

4.02 Trigonometry values

Spreadsheet (1):

4.02 Trigonometry values

Presentation (1):

4.07 Applications of triangles

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- investigate the sine, cosine and tangent ratios in similar right-angled triangles
- use trigonometry to solve problems involving right-angled triangles
- find unknown sides and angles in right-angled triangles, where the angle is measured in degrees and minutes
- solve trigonometry problems involving bearings and angles of elevation and depression

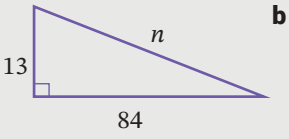
SkillCheck ANSWERS ON P. 617

1 Solve each equation.

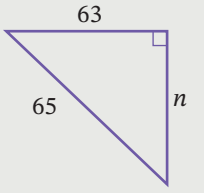
a $\frac{x}{5} = 7$ **b** $\frac{h}{4} = 8.3$ **c** $\frac{45}{y} = 9$

2 For each triangle, find the value of n.

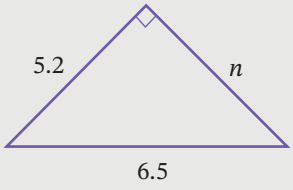
a



b



c



3 Round each time to the nearest hour.

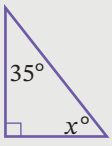
a 8 h 18 min **b** 3 h 45 min **c** 1 h 30 min

4 Convert each time to hours and minutes.

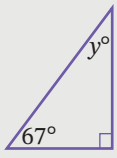
a 4.7 h **b** 2.25 h **c** 6.85 h

5 Find the value of each variable.

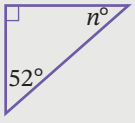
a



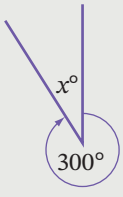
b



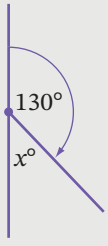
c



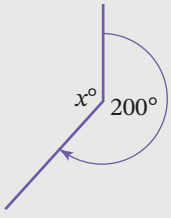
d



e



f



Quiz
SkillCheck 4

Videos
Measuring the Earth
The tunnel of Samos
Distance to the Sun and Moon

4.01 The sides of a right-angled triangle

The 3 sides of a **right-angled triangle** have special names. These names depend on the position of a given angle in the triangle.

- The **hypotenuse** is the longest side and is always opposite the right angle
- The **opposite side** directly faces the given angle
- The **adjacent side** runs from the given angle to the right angle

We have already learnt about the hypotenuse in Pythagoras' theorem.

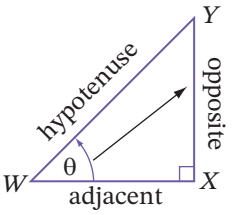
Adjacent means 'next to'.

In this diagram, the marked angle $\angle W$ has also been labelled with the Greek letter, θ, ('theta').

The hypotenuse is WY.

The opposite side is XY.

The adjacent side is WX.

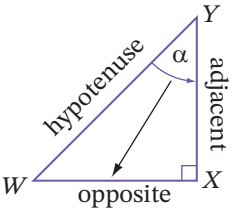


In this second diagram, the marked angle $\angle Y$ has also been labelled with the Greek letter α ('alpha'), so the opposite and adjacent sides are different.

The hypotenuse is WY.

The opposite side is WX.

The adjacent side is XY.

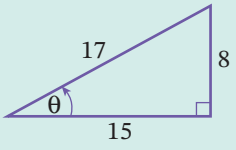


Video
The sides of a right-angled triangle

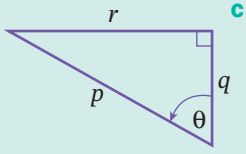
Example 1

For each triangle, name the hypotenuse, opposite and adjacent sides for angle θ.

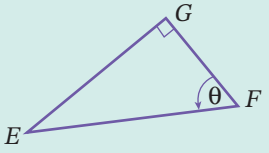
a



b



c



Solution

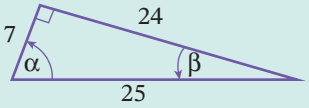
a Hypotenuse is 17.
Opposite side is 8.
Adjacent side is 15.

b Hypotenuse is p
Opposite side is r.
Adjacent side is q.

c Hypotenuse is EF.
Opposite side is EG.
Adjacent side is FG.

Example 2

For angles α and β, name the adjacent side.



Solution

For α, the adjacent side is 7.

For β, the adjacent side is 24.

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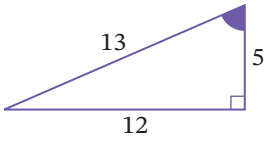
EXERCISE 4.01 ANSWERS ON P. 617

The sides of a right-angled triangle U F C

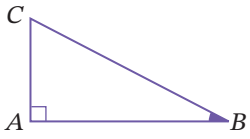
EXAMPLE 1

1 For the marked angle in each triangle, name the hypotenuse, opposite and adjacent sides. C

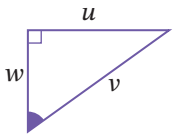
a



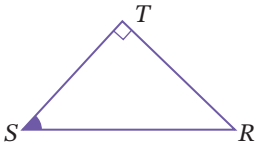
b



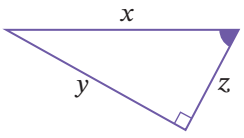
c



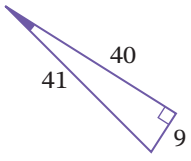
d



e



f



2 For $\triangle LKM$, name the angle:

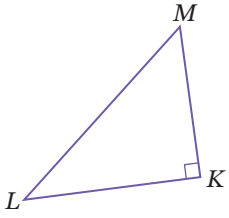
a opposite the hypotenuse

b opposite side KM

c opposite side LK

d adjacent to side KM

e adjacent to side LK.



3 Which one of these statements is **false** about $\triangle TSU$?

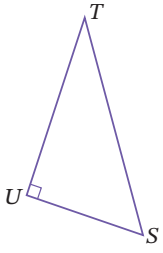
Select the correct answer **A**, **B**, **C** or **D**.

A The adjacent side to $\angle S$ is US.

B The adjacent side to $\angle T$ is TU.

C The hypotenuse is UT.

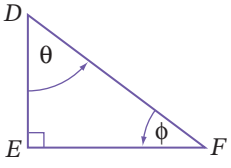
D The opposite side to $\angle T$ is US.



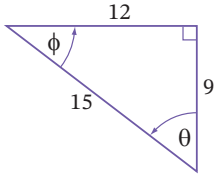
EXAMPLE 2

4 For each triangle, find the opposite sides for angles θ ('theta') and ϕ ('phi').

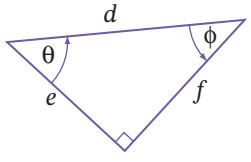
a



b

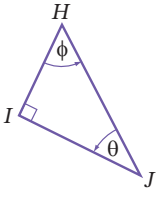


c

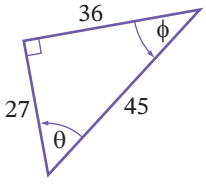


5 For each triangle, find the adjacent sides for angles θ and ϕ.

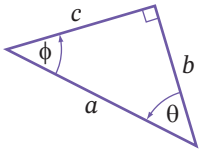
a



b



c



Foundation | Standard | Complex

6 Which side of a right-angled triangle is fixed and does not depend on a given angle? Select **A**, **B**, **C** or **D**.

A adjacent **B** hypotenuse **C** opposite **D** shortest

7 Given each description of a right-angled triangle, sketch the triangle with the correctly-labelled vertices and angle. C

a $\triangle ABC$ has hypotenuse AB and side AC opposite angle θ

b $\triangle XYZ$ has hypotenuse YZ and side XZ adjacent to angle α

c $\triangle PRQ$ has side RQ opposite $\angle P$ and adjacent to $\angle R$

d $\triangle DEF$ is right-angled at E, with the opposite and adjacent sides of $\angle D$ equal

Did you know?

The Greek alpha-bet

Here are 8 letters (in lower-case and capitals) from the Greek alphabet:

α, A alpha β, B beta γ, Γ gamma δ, $\triangle$ delta

θ, Θ theta π, Π pi σ, Σ sigma ω, Ω omega

Greek letters are commonly used as variables, particularly in geometry and trigonometry.

1 **Find how many letters there are in the Greek alphabet, and name each one.**

2 **Compare the Greek alphabet with our Roman alphabet.**

3 **Can you see where the word alphabet comes from? Explain how it originated.**

4.02 The trigonometric ratios

Puzzle Trigonometry match-up

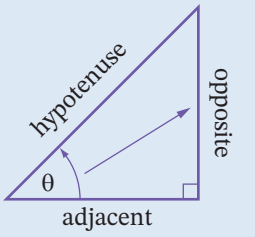
Technology Trigonometry values

Spreadsheet Trigonometry values

There are 3 special fractions called **trigonometric ratios** that relate the lengths of 2 sides of a right-angled triangle: **sine**, **cosine** and **tangent**.

The trigonometric ratios

Ratio	Abbreviation	Meaning
sine	sin	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
cosine	cos	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
tangent	tan	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$

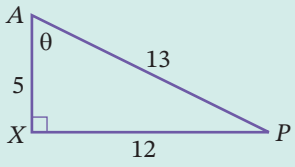


Foundation Standard Complex

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Example 3

In $\triangle AXP$, find $\sin\theta$, $\cos\theta$ and $\tan\theta$.



Solution

For angle θ, opposite = 12, adjacent = 5, hypotenuse = 13.

$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{12}{13}$ $\quad$ $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{5}{13}$

$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{12}{5}$

A useful **mnemonic** (memory aid) for remembering the 3 ratios is to look at the initials of the words in the ratios:

$\sin = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\text{O}}{\text{H}}$ SOH

$\cos = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\text{A}}{\text{H}}$ CAH

$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{\text{O}}{\text{A}}$ TOA

If you remember **SOH-CAH-TOA** (pronounced 'so-car-toe-ah'), then you can remember the ratios for sin, cos and tan. Some students also learn a phrase where the first letter of each word follows the SOH-CAH-TOA sequence, for example, 'Sun Over Head Caused A Huge Tan On Arms'. Find your own mnemonic for the 3 ratios.

Example 4

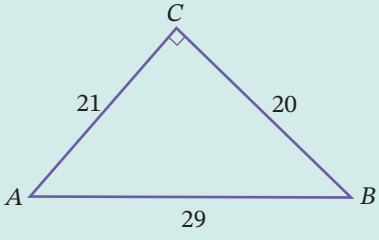
For the triangle shown, find:

a $\sin A$

b $\cos B$

c $\tan A$

d $\sin B$



Solution

For angle A, opposite = 20, adjacent = 21, hypotenuse = 29

For angle B, opposite = 21, adjacent = 20, hypotenuse = 29

a $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{20}{29}$ $\quad$ **b** $\cos B = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{20}{29}$

c $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{20}{21}$ $\quad$ **d** $\sin B = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{21}{29}$

4.02

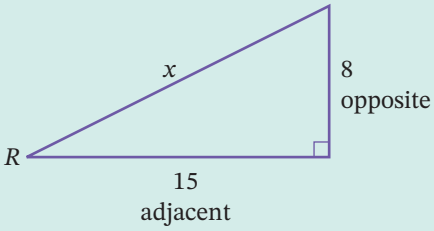
Video The trigonometric ratios

Example 5

If $\tan R = \frac{8}{15}$, find the value of sin R and cos R.

Solution

Since $\tan R = \frac{\text{opposite}}{\text{adjacent}} = \frac{8}{15}$, draw a right-angled triangle that has an angle R with opposite side 8 and adjacent side 15. Let x be the length of the hypotenuse.



Find x using Pythagoras' theorem.

$x^2 = 8^2 + 15^2$

$= 289$

$x = \sqrt{289}$

$= 17$

$\therefore \sin R = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}$

$\cos R = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{15}{17}$

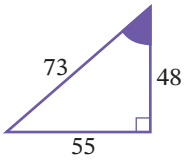
EXERCISE 4.02 ANSWERS ON P. 618

The trigonometric ratios U F R C

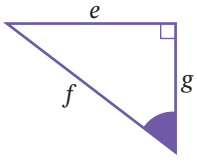
EXAMPLE 3

1 For each marked angle, find the sine, cosine and tangent ratios. C

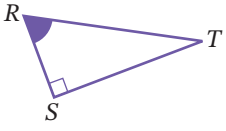
a



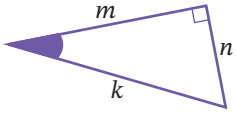
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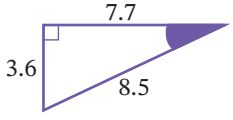
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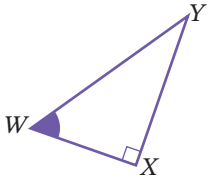
d



e



f

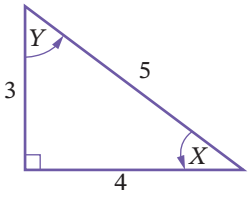


EXAMPLE 4

2 For the triangle shown, find:

a cos X **b** tan Y

c sin X **d** sin Y



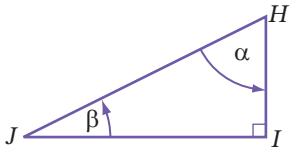
Foundation Standard Complex

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3 Complete each statement below with the correct angle (α or β).

a $\sin\square = \frac{IJ}{HJ}$ **b** $\sin\square = \frac{HI}{HJ}$ **c** $\cos\square = \frac{IJ}{HJ}$

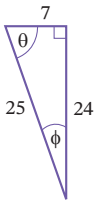
d $\cos\square = \frac{HI}{HJ}$ **e** $\tan\square = \frac{IJ}{HI}$ **f** $\tan\square = \frac{HI}{IJ}$



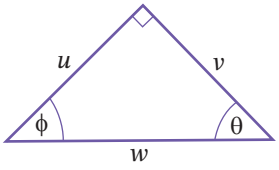
4 For each triangle below, find:

i $\tan\theta$ **ii** $\cos\theta$ **iii** $\cos\phi$ **iv** $\tan\phi$ ϕ = 'phi'

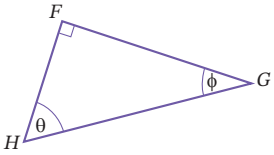
a



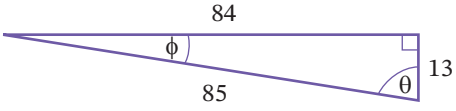
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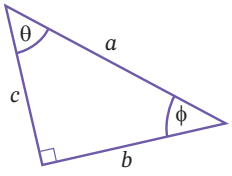
c



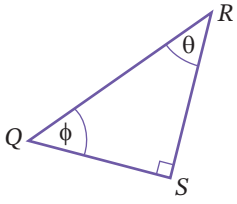
d



e



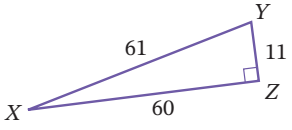
f



5 For each fraction, write a correct trigonometric ratio involving angle X or Y in the triangle. C

a $\frac{60}{11}$ **b** $\frac{11}{60}$

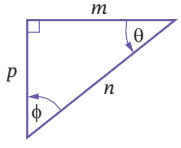
c $\frac{11}{61}$ **d** $\frac{60}{61}$



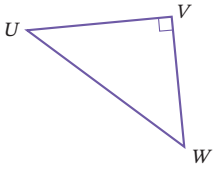
6 Which ratio is equal to $\frac{p}{m}$? Select the correct answer **A**, **B**, **C** or **D**.

A $\cos\theta$ **B** $\cos\phi$

C $\tan\theta$ **D** $\tan\phi$



7 Which statement is true for this triangle?
Select **A**, **B**, **C** or **D**. R C



A $\sin U = \cos W$ **B** $\tan U = \sin W$

C $\cos U = \tan W$ **D** $\tan U = \tan W$

8 Sketch a right-angled triangle for each trigonometric ratio, then use Pythagoras' theorem to find the length of the unknown side and the other 2 trigonometric ratios for the same angle.

a $\tan A = \frac{5}{12}$ **b** $\sin B = \frac{3}{5}$ **c** $\cos X = \frac{9}{41}$ **d** $\sin Y = \frac{7}{25}$

4.02

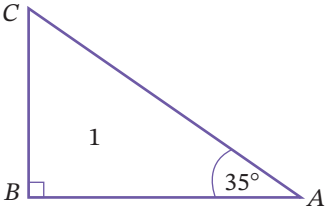
EXAMPLE 5

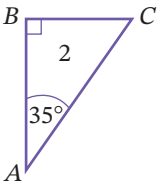
Foundation Standard Complex

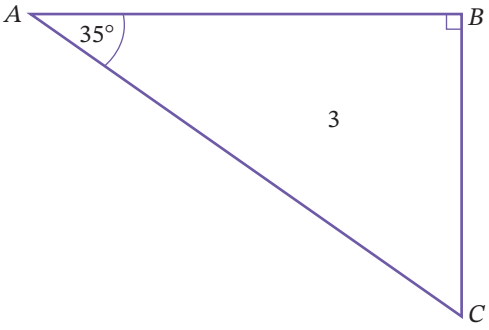
4.03 Similar right-angled triangles

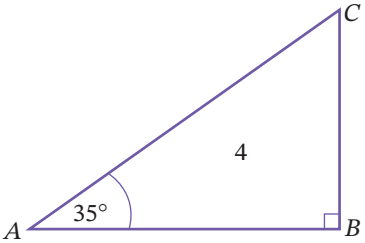
Worksheet
Investigating the tangent ratio

In each right-angled triangle below, $\angle A = 35°$.









Furthermore, because $\angle B = 90°$, $\angle C = 180° - 90° - 35° = 55°$ because of the angle sum of a triangle.

These 4 triangles are called **similar triangles** because they have the same shape but are not the same size. In fact, they are enlargements or reductions of one another.

Example 6

For each triangle above, measure the length of each side (correct to the nearest mm) and then calculate $\sin A$, $\cos A$ and $\tan A$ as decimals (correct to 2 decimal places).

Solution

	Side length (mm)			Trigonometric ratio		
	BC (opp)	AB (adj)	AC (hyp)	$\sin A = \frac{BC}{AC}$	$\cos A = \frac{AB}{AC}$	$\tan A = \frac{BC}{AB}$
1	25	36	44	25 ÷ 44 ≈ 0.57	36 ÷ 44 ≈ 0.82	25 ÷ 36 ≈ 0.69
2	14	20	24	0.58	0.83	0.70
3	40	57	69	0.59	0.83	0.70
4	29	41	50	0.58	0.82	0.71

Similar triangles have matching equal angles. We will learn more about them in Chapter 13, *Congruent and similar figures*.

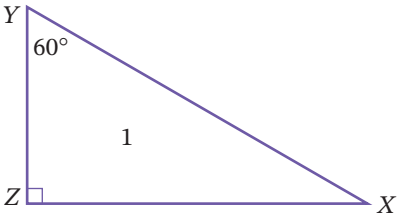
Note: See **Technology: Similar right-angled triangles** after Exercise 4.03 for a dynamic geometry activity based on this example.

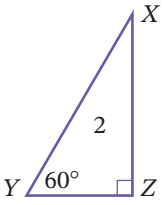
Similar right-angled triangles U F R C

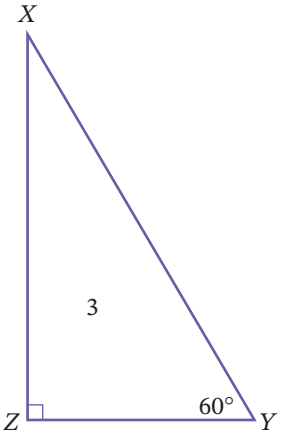
EXAMPLE 6

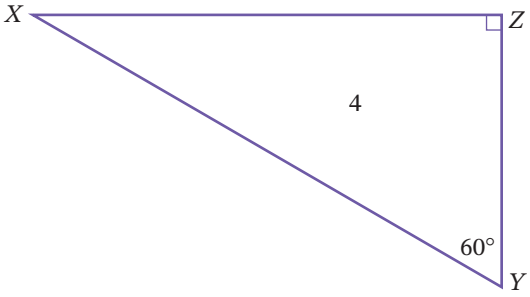
4.03

1 a For each similar right-angled triangle below, measure the length of each side (correct to the nearest mm) and then calculate sin Y, cos Y and tan Y as decimals (correct to 2 decimal places). Copy and complete the table. R C.









	Side length (mm)			Trigonometric ratio		
	XZ (opp)	ZY (adj)	XY (hyp)	$\sin Y = \frac{XZ}{XY}$	$\cos Y = \frac{ZY}{XY}$	$\tan Y = \frac{XZ}{ZY}$
1						
2						
3						
4						

b What do you notice about the value of sin Y for all 4 similar right-angled triangles?

c Use your calculator to evaluate sin 60° by pressing sin 60 =.
What do you notice about your answer?

d What do you notice about the value of cos Y for all 4 similar triangles?

e Use your calculator to evaluate cos 60°. What do you notice about your answer?

f What do you notice about the value of tan Y for all 4 similar triangles?

g Use your calculator to evaluate tan 60°. What do you notice about your answer?

2 **a** Draw 4 similar right-angled triangles that have an angle of 48°, measure the length of each side (correct to the nearest mm) and then calculate sin 48°, cos 48° and tan 48° as decimals (correct to 2 decimal places). Copy and complete the table. R C

	Side length (mm)			Trigonometric ratio		
	Opp	Adj	Hyp	$\sin 48° = \frac{\text{opp}}{\text{hyp}}$	$\cos 48° = \frac{\text{adj}}{\text{hyp}}$	$\tan 48° = \frac{\text{opp}}{\text{adj}}$
1						
2						
3						
4						

b Examine the value of sin 48° for all 4 similar triangles, then evaluate sin 48° on a calculator. What do you notice?

c Examine the values of cos 48°, then evaluate cos 48° on a calculator.

d Examine the values of tan 48°, then evaluate tan 48° on a calculator.

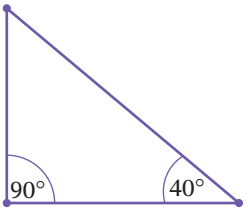
3 For each trigonometric ratio, draw a large right-angled triangle with the given angle, then by measurement and calculation, find the value of the ratio, correct to 3 decimal places. Compare your answer to the calculator's answer. R C

a tan 55° **b** cos 39° **c** sin 67° **d** cos 21°

Technology

Similar right-angled triangles

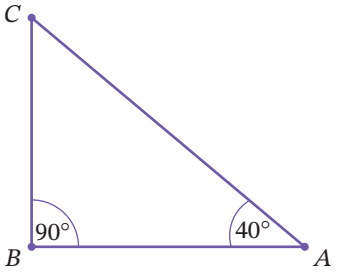
We will use dynamic geometry software to measure and calculate trigonometric ratios.



1 **a** Before you start, set the measuring tool to measure angles to degrees.

b By constructing intervals, draw a right-angled triangle. Then use the measuring tool to measure the right angle and another angle.

c Now keeping the triangle right-angled, move any of the other 2 vertices so that one angle is 40°.



2 Draw 2 more similar right-angled triangles with a 40° angle.

3 Label the vertices of each triangle as A, B and C.

Make sure $\angle A = 40°$ and $\angle B = 90°$.

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4 **a** Copy this table.

	Side length (mm)			Trigonometric ratio		
	BC (opp)	AB (adj)	AC (hyp)	$\sin A = \frac{BC}{AC}$	$\cos A = \frac{AB}{AC}$	$\tan A = \frac{BC}{AB}$
1						
2						
3						

b Use the measurement tool to measure the sides of each triangle and write them in the table.

c Then, for each triangle, calculate the trigonometric ratios, to 2 decimal places.

d What do you notice about the values of each ratio for all of the triangles?

Trigonometry on a calculator 4.04

In the previous section, we discovered that for any particular angle, the sine, cosine and tangent ratios stay constant (the same) for all right-angled triangles with that angle. For example, $\sin 35° \approx 0.57$ always, no matter what size the similar right-angled triangle.

Puzzle
Trigonometry squaresaw

Worksheet
Trigonometric calculations

Constancy of the sine, cosine and tangent ratios

For any given angle, the values of the sine, cosine and tangent ratios are constant.

This means that the value of a trigonometric ratio can be easily found on a calculator rather than through constructing and measuring triangles.

Angles are measured in **degrees**, but one degree can be subdivided into 60 **minutes**.

One minute can be further subdivided into 60 **seconds**. The abbreviations for minutes and seconds are shown below.

Degrees and minutes

$1° = 60'$ (1 degree = 60 minutes)

$1' = 60''$ (1 minute = 60 seconds)

For example, an angle size of $48° 35' 56''$ is 48 degrees, 35 minutes and 56 seconds, about halfway between 48° and 49°.

To enter degrees, minutes and seconds into a scientific calculator, use the °,," or D°M'S (Degrees-Minutes-Seconds) key.

Videos
Trigonometry on a calculator

Example 7

Evaluate each expression correct to 2 decimal places.

a $\sin 46°$ **b** $\tan 57.4$ **c** $4 \cos 20°$

d $68.3 \sin 38°25'$ **e** $\dfrac{23}{\cos 18°50'}$

Solution

Make sure that your calculator is in the degrees mode (D or DEG) or your answer will be incorrect.

a $\sin 46° = 0.71933...$
≈ 0.72

On calculator: sin 46 =

b $\tan 57.4° = 1.56365...$

This angle is 57.4° (a decimal), not 57°4′.

≈ 1.56

On calculator: tan 57.4 =

c $4 \cos 20° = 3.75877...$

This means $4 \times \cos 20°$.

≈ 3.76

On calculator: 4 cos 20 =

d $68.3 \sin 38°25' = 42.43996...$
≈ 42.44

On calculator: 68.3 sin 38 °’” 25 °’” =

e $\dfrac{23}{\cos 18°50'} = 24.30103...$
≈ 24.30

On calculator: 23 ÷ cos 18 °’” 50 °’” =

Rounding angle sizes

When rounding an angle to the nearest degree or minute, use 30 as the halfway mark.

Video
Rounding angles

Example 8

Round each angle correct to the nearest degree.

a 73° 27′ **b** 9° 41′ **c** 42° 30′ **d** 128° 29′ 47″

Solution

a $73°27' \approx 73°$ — $27' < 30'$, so round down.

b $9°41' \approx 10°$ — $41' \geq 30'$, so round up.

c $42°\ 30' \approx 43°$ — $30' \geq 30'$, so round up.

d $128°\ 29'\ 47'' \approx 128°$ — $29' < 30'$, so round down.

Example 9

Round each angle correct to the nearest minute.

a 33°53′30″ **b** 44°15′40″

Solution

a 33°53′ 30″ ≈ 33°54′ 30″ ≥ 30″, so round up.

b 44°15′40″ ≈ 44°16″ 40″ ≥ 30″, so round up.

Video
Rounding angles

Example 10

Convert each angle size to degrees and minutes, correct to the nearest minute.

a 82.5° **b** 60.81°

Solution

a 82.5° = 82° 30′ On a calculator: 82.5 = °′″

b 60.81° = 60° 48′ 36″
≈ 60° 49′ On a calculator: 60.81 = °′″

EXERCISE 4.04 ANSWERS ON P. 618

Trigonometry on a calculator U F

1 Evaluate each expression, correct to 2 decimal places.

EXAMPLE 7

a	tan 84°	**b**	cos 15°	**c**	tan 47°	**d**	sin 33°
e	sin 77°	**f**	cos 60.1°	**g**	tan 39.55°	**h**	cos 18°
i	8 tan 75°	**j**	14 sin 56°	**k**	12 ÷ tan 20°	**l**	$\frac{7}{\sin 43°}$
m	50 × sin 70° 34′	**n**	66.2 cos 81° 42′	**o**	18.53 sin 11° 8′	**p**	$\frac{27}{\cos 35°22′}$
q	$\frac{44.5}{\tan 65°58′}$	**r**	$\frac{200}{\sin 54°47′}$	**s**	24.1 ÷ tan 63° 2′	**t**	$\frac{15.7}{\cos 21°8′}$

2 Round each angle size correct to the nearest degree.

EXAMPLE 8

a	27° 54′	**b**	40° 30′	**c**	19° 18′	**d**	33° 7′ 25″
e	33° 41′ 5″	**f**	56.4°	**g**	29.75°	**h**	44° 18′

3 Round each angle size correct to the nearest minute.

EXAMPLE 9

a	68° 39′ 42″	**b**	54° 22′ 21″	**c**	24° 46′ 30″	**d**	18° 30′ 27″
e	9° 10′ 55″	**f**	47° 59′ 9.5″	**g**	3° 45′ 35″	**h**	57° 10′ 29″

4 Convert each angle size to degrees and minutes, correct to the nearest minute.

EXAMPLE 10

a	55.5°	**b**	14.15°	**c**	72.38°	**d**	33.77°
e	66.41°	**f**	7.875°	**g**	28.123°	**h**	31.046°
i	34.45°	**j**	71.087°	**k**	5.4829°	**l**	69.4545°

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5 By guess-and-checking with your calculator, find the angle size, θ (to the nearest degree), that gives each value.

a $\sin\theta = 0.7880$ **b** $\tan\theta = 0.2493$ **c** $\tan\theta = 1.2799$

d $\cos\theta = 0.5$ **e** $\sin\theta = 0.5446$ **f** $\cos\theta = 0.8829$

g $\tan\theta = 0.7265$ **h** $\sin\theta = 0.9998$

Quiz
Mental skills 4

Mental skills 4: Maths without calculators ANSWERS ON P. 619

Estimating answers

A quick way of estimating an answer is to round each number in the calculation.

1 Study each example.

a $631 + 280 + 51 + 43 + 96$
$\approx 600 + 300 + 50 + 40 + 100$
$= (600 + 300 + 100) + (50 + 40)$
$= 1000 + 90$
$= 1090$ (Actual answer = 1101)

b $55 + 132 - 34 + 17 - 78$
$\approx 60 + 130 - 30 + 20 - 80$
$= (60 + 20 - 80) + (130 - 30)$
$= 0 + 100$
$= 100$ (Actual answer = 92)

c $78 \times 7 \approx 80 \times 7$
$= 560$ (Actual answer = 546)

d $510 \div 24 \approx 500 \div 20$
$= 50 \div 2$
$= 25$ (Actual answer = 21.25)

2 Now estimate each answer.

a $27 + 11 + 87 + 142 + 64$ **b** $55 + 34 - 22 - 46 + 136$

c $684 + 903$ **d** $35 + 81 + 110 + 22 + 7$

e $517 - 96$ **f** $210 - 38 - 71 + 151 - 49$

g $766 - 353$ **h** 367×2

i 83×81 **j** 984×16

k $828 \div 3$ **l** $507 \div 7$

3 Study each example involving decimals.

a $20.91 - 11.3 + 2.5$
$\approx 21 - 11 + 3$
$= 13$ (Exact answer = 12.11)

b $4.78 \times 19.2 \approx 5 \times 20$
$= 100$ (Exact answer = 91.776)

c $75.13 \div 8.4 \approx 75 \div 8$
$< 80 \div 8$
< 10
≈ 9 (Exact answer = 8.944...)

d $\dfrac{37.6 + 9.3}{41.2 - 12.7} \approx \dfrac{38 + 9}{40 - 13}$
$= \dfrac{47}{27}$
$\approx \dfrac{50}{30}$
≈ 1.6 (Exact answer = 1.645…)

4 Now estimate each answer.

a $3.75 + 9.381 + 4.6 + 10.5$ **b** $14.807 + 6.6 - 7.22$

c 18.47×9.61 **d** 4.27×97.6

e $\dfrac{11.07 + 18.4}{12.2}$ **f** $\dfrac{38.18}{17.2 - 9.6}$

g $54.75 - 18.6 - 14.4$ **h** $\dfrac{18.46 \times 4.9}{39.72 - 15.2}$

i $62.13 \div 10.7$ **j** $(4.89)^2$

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Did you know?

Degrees, minutes and seconds

We use base 10 systems in number, measurement and currency: there are 100 centimetres in a metre, 1000 grams in a kilogram and 100 cents in a dollar. So why are there 360° in a revolution, 60 minutes in a degree and 60 seconds in a minute?

In 2000 BCE, the Babylonians used a base 60 or sexagesimal system of numeration, because 60 is a rounder, more convenient number than 10. This is because 60 has more factors and is divisible by 3, 4 and 6. Furthermore, $6 \times 60 = 360$, which was the Babylonian approximation for the number of days in a year, so that each day the Earth would travel 1° around the Sun.

As measuring devices and calculations required greater precision, each degree was subdivided into 60 equal parts called minutes, and these were further divided into 60 parts called seconds. This level of accuracy is essential in navigation and mapping.

1 **A minute is a 'small' part of a degree. Investigate how an alternative meaning (and pronunciation) of 'minute' is 'tiny'.**

2 **A second is the 'second' subdivision of a degree. Explain how there are 2 different meanings of 'second'.**

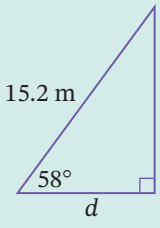
Finding an unknown side 4.05

Since the trigonometric ratio of any angle is a constant number, we can use it to calculate the length of an unknown side in a right-angled triangle if one other side is known. We need to select the correct ratio that links the given angle to the unknown side and known side.

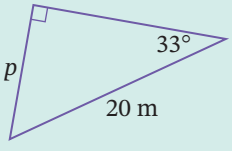
Example 11

Find the value of each variable, correct to 2 decimal places.

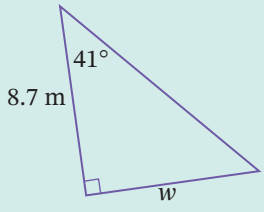
a



b



c



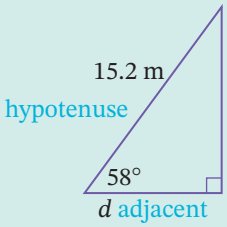
Solution

a SOH, CAH OR TOA?

The marked sides related to 58° are the adjacent side (A) and the hypotenuse (H), so use cos.

$$\cos 58° = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$= \frac{d}{15.2}$$

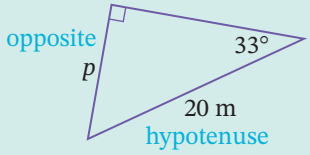


Videos
Finding an unknown side
Trigonometry

$15.2 \times \cos 58° = 15.2 \times \frac{d}{15.2}$ Multiplying both sides by 15.2

$15.2 \cos 58° = d$
$d = 15.2 \cos 58°$
$= 8.05477...$
$= 8.05$
From the diagram, a length of 8.05 m looks reasonable.

b SOH, CAH OR TOA?
The marked sides related to 33° are the opposite side (O) and the hypotenuse (H), so use sin.



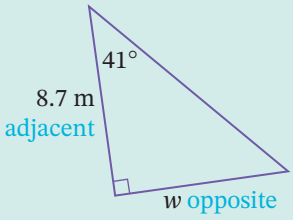
$\sin 33° = \frac{\text{opposite}}{\text{hypotenuse}}$

$= \frac{p}{20}$

$20 \times \sin 33° = 20 \times \frac{p}{20}$

$20 \sin 33° = p$
$p = 20 \sin 33°$
$= 10.8927...$
$= 10.89$
From the diagram, a length of 10.89 m looks reasonable.

c SOH, CAH OR TOA?
The marked sides related to 41° are the opposite side (O) and the adjacent side (A), so use tan.



$\tan 41° = \frac{\text{opposite}}{\text{adjacent}}$

$= \frac{w}{8.7}$

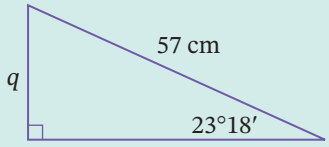
$w = 8.7 \tan 41°$
$= 7.5627...$
$= 7.56$
From the diagram, a length of 7.56 m looks reasonable.

Finding an unknown side in a right-angled triangle

1 Identify the 2 labelled sides and decide whether to use sin, cos or tan
2 Write an equation using the ratio, the given angle and the variable
3 Solve the equation to find the value of the variable

Example 12

Find the value of q, correct to the nearest centimetre.



Solution

q is opposite 23° 18′, 57 cm is the hypotenuse, so use sin.

$$\sin 23° 18' = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin 23°18' = \frac{q}{57}$$

$$q = 57 \sin 23° 18'$$
$$= 22.5460...$$
$$= 23 \text{ cm}$$

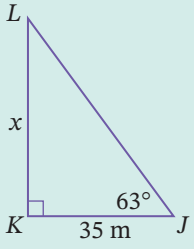
From the diagram, a length of 23 cm looks reasonable.

Example 13

$\triangle JKL$ is right-angled at K, $JK = 35$ m and $\angle J = 63°$. Find the length of LK, correct to the nearest metre.

Solution

Draw a diagram.



Let the length of LK be x.

x is opposite, 35 m is adjacent, so use tan.

$$\tan 63° = \frac{x}{35}$$

$$x = 35 \tan 63°$$
$$= 68.6913...$$
$$LK \approx 69 \text{ m}$$

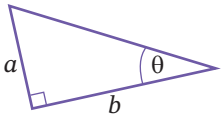
From the diagram, a length of 69 m looks reasonable.

EXERCISE 4.05 ANSWERS ON P. 619

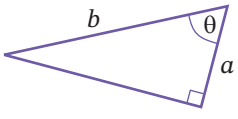
Finding an unknown side U F PS

1 For each triangle, which trigonometric ratio ($\sin\theta$, $\cos\theta$ or $\tan\theta$) is equal to $\frac{a}{b}$?

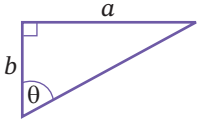
a



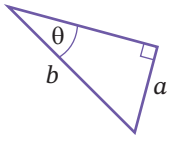
b



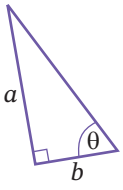
c



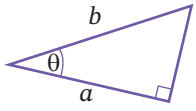
d



e



f

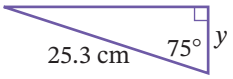


Foundation | Standard | Complex

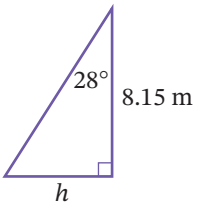
EXAMPLE 11

2 Find the value of the variable in each triangle, correct to 2 decimal places.

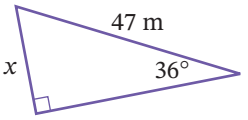
a



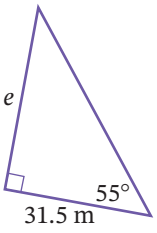
b



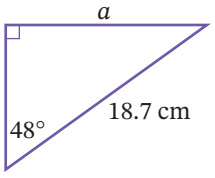
c



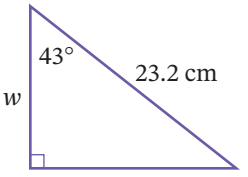
d



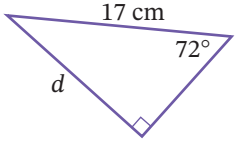
e



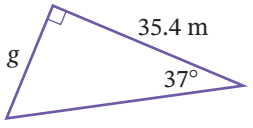
f



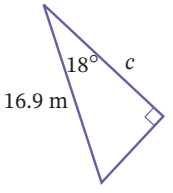
g



h



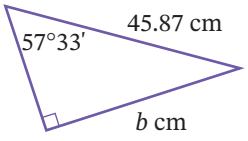
i



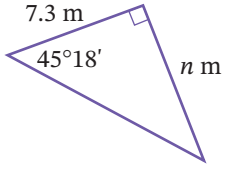
EXAMPLE 12

3 Find the value of the variable in each triangle, correct to 2 decimal places.

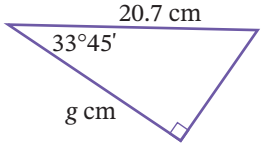
a



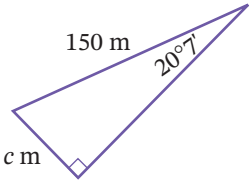
b



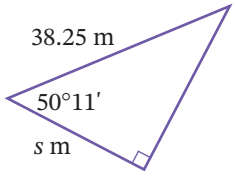
c



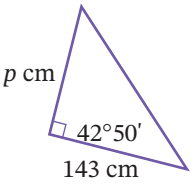
d



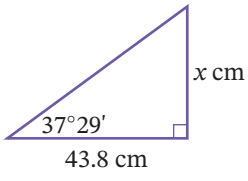
e



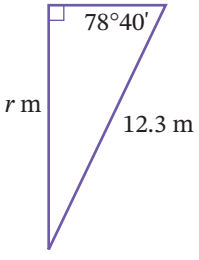
f



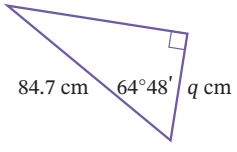
g



h



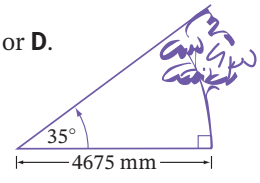
i



4 What is the height of this tree? Select the correct answer **A**, **B**, **C** or **D**.

A 2.68 m **B** 3.27 m

C 3.83 m **D** 6.68 m

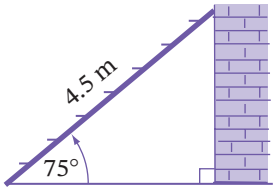


Foundation Standard Complex

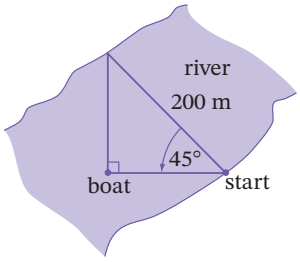
9780170479585

5 Find each length or distance, correct to one decimal place. PS

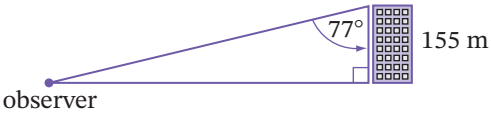
a How high the ladder reaches up the wall.



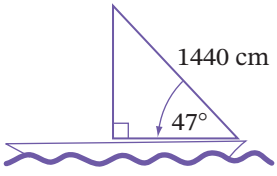
b The distance between the boat and the start.



c The distance from the observer to the base of the building.



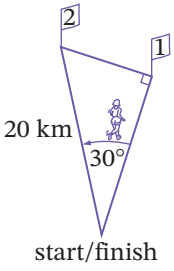
d The height of the boat's mast.



e The distance between:

i checkpoints 1 and 2

ii checkpoint 1 and the start.



EXAMPLE 13

6 $\triangle ABC$ is right-angled at B, $AC = 14.8$ m and $\angle C = 56°$. Find the length of side AB, correct to one decimal place.

7 $\triangle MNR$ is right-angled at M, $MN = 19$ cm and $\angle N = 27°$. Find the length of MR, correct to the nearest centimetre.

8 In $\triangle XYW$, $\angle X = 90°$, $\angle Y = 43.7°$ and $WY = 8.34$ m. Find the length of XW, correct to 2 decimal places.

9 $\triangle AHK$ is right-angled at K, $\angle H = 76°$ and $AH = 13.9$ m. Find the length of HK, correct to one decimal place.

10 A tree casts a shadow 20 m long. If the Sun's rays meet the ground at 25°, find the height of the tree, correct to the nearest cm. PS

11 A 6 m ladder is placed against a pole. If the ladder makes an angle of 17° with the pole, how far up the pole does the ladder reach? Answer to the nearest mm. PS

12 A boat is anchored by a rope 5.5 m long. If the anchor rope makes an angle of 23° with the vertical, calculate the depth of the water (correct to one decimal place). PS

Foundation | Standard | Complex

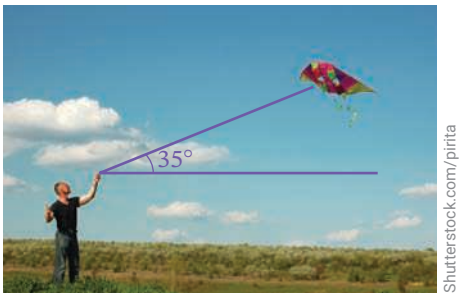
13 A rectangular gate has a diagonal brace that makes an angle of 60° with the bottom of the gate. If the length of the diagonal brace is 1860 mm, calculate the height of the gate. Select **A**, **B**, **C** or **D**. PS

A 2148 mm **B** 930 mm **C** 1611 mm **D** 3221 mm

14 Tayyab is flying a kite that is attached to a string 155 m long. The string makes an angle of 35° to the horizontal. How high, to the nearest metre, is the kite above Tayyab's hand? Select **A**, **B**, **C** or **D**. PS

A 78 m **B** 89 m

C 109 m **D** 127 m



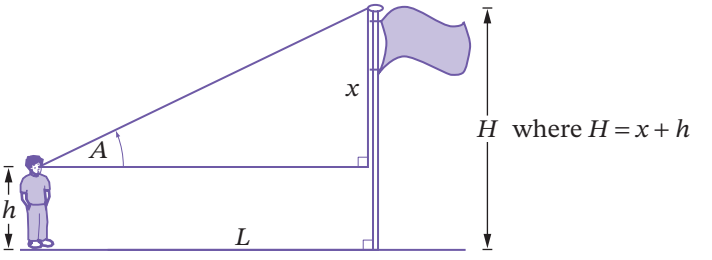
Shutterstock.com/pirita

Investigation

Calculating the height of an object

You will need: tape measure or trundle wheel, a clinometer (or protractor) to measure the angle.

Trigonometry can be used to find the heights of buildings, flagpoles and trees without actually measuring them. This can be done by measuring the distance along the ground from the base of the object to a person. The person then measures the angle to the top of the object. For example, the height of a flagpole can be calculated using the set-up shown in the diagram.



- h is the eye height of the person who measures the angle, A, to the top of the flagpole
- L is the distance the person is from the base of the flagpole
- x is the height of the flagpole above the person's eye height
- $H = x + h$ is the height of the flagpole above the ground.

Foundation Standard Complex

9780170479585

1. Select a tall object outside to measure.
2. Work with a partner to measure (in cm) the distance, L, along the ground, the height, h, of the person, and the angle (in degrees) to the top of the object. Copy the table below and record your information in the first row.

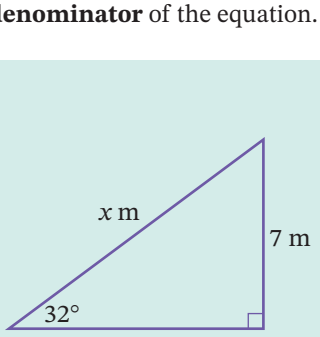
Distance, L (cm)	Angle, $A°$	Height of person, h (cm)	Calculated height, x cm	Height of flagpole, H cm

3. Use the tan ratio to calculate the value of x, correct to the nearest whole number.
4. Hence find H, the height of the kite, correct to the nearest centimetre. Write your answers in the table.
5. Repeat the measurements and calculations three more times from different positions, with different persons measuring the angle. This will help to improve the accuracy of your results and minimise errors. Write your results in the table.
6. Did you find similar values for H? Do they seem reasonable for the height of the object?
7. Calculate the average value for H.

Finding more unknown sides

4.06

In the following examples, the unknown appears in the **denominator** of the equation.



Example 14

Find the value of x, correct to one decimal place.

Solution

7 m is the opposite side, x m is the hypotenuse, so use sin.

$\sin 32° = \frac{7}{x}$

Note that the variable x appears in the denominator of the equation

$\sin 32° \times x = \frac{7}{x} \times x$

$x \sin 32° = 7$

$\frac{x \sin 32°}{\sin 32°} = \frac{7}{\sin 32°}$

$x = \frac{7}{\sin 32°}$

$= 13.2095...$

≈ 13.2

Note that when the unknown appears in the denominator of an equation, it can swap positions with the trigonometric ratio, so that $\sin 32° = \frac{7}{x}$ becomes $x = \frac{7}{\sin 32°}$.

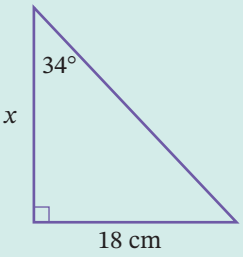
Worksheet
Finding an unknown side

Puzzle
Trigonometry equations 1

Video
Finding the hypotenuse

Example 15

Find the length of x, correct to 2 decimal places.



Solution

18 cm is the opposite side, x is the adjacent side, so use tan.

$\tan 34° = \frac{18}{x}$ — x appears in the denominator

$x = \frac{18}{\tan 34°}$ — Swap the position of x with tan 34°

$= 26.6860...$

≈ 26.69 cm

Alternative method

To avoid having x in the denominator, we could use tan with the 3rd angle of the triangle.

3rd angle $= 180° - 90° - 34° = 56°$

$\tan 56° = \frac{x}{18}$

$x = 18 \tan 56°$

$= 26.6860...$

≈ 26.69 cm

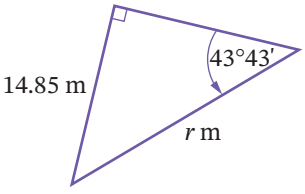
EXERCISE 4.06 ANSWERS ON P. 619

Finding more unknown sides U F PS

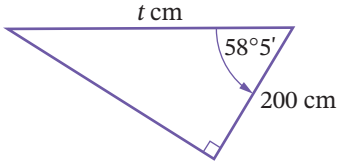
EXAMPLE 14

1 Find the value of each variable, correct to one decimal place.

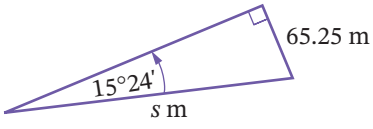
a



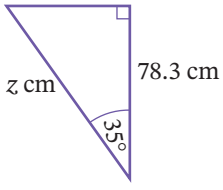
b



c



d



2 $\triangle XYZ$ is right-angled at Z, $ZY = 230$ mm and $\angle Y = 45°$. Find the length of XY, correct to the nearest millimetre.

3 In $\triangle KLW$, $\angle L = 90°$, $KL = 12$ m and $\angle W = 75.2°$. Find KW, correct to one decimal place.

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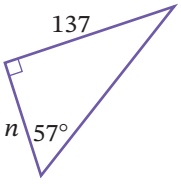
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EXAMPLE 15

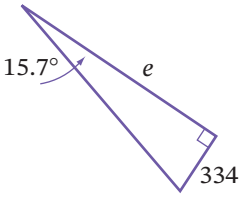
4.06

4 Find the value of each variable, correct to 2 decimal places.

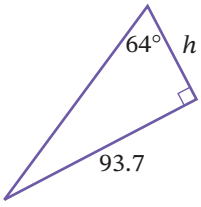
a



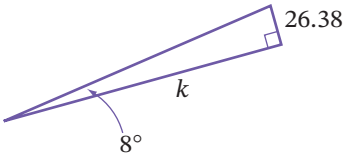
b



c



d

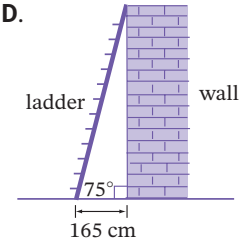


5 $\triangle CDE$ is right-angled at D, $\angle E = 36°$ and $CD = 5$ m. Find the length of side DE, correct to 2 decimal places.

6 Find the length of this ladder. Select the correct answer **A**, **B**, **C** or **D**.

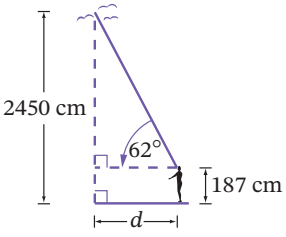
A 159 cm **B** 171 cm

C 243 cm **D** 638 cm

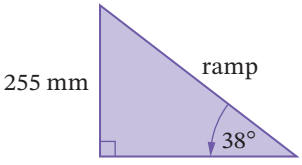


7 Find each length or distance, correct to one decimal place. PS

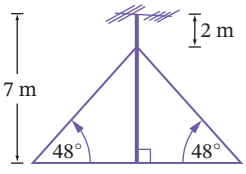
a How far the person is from being directly under the birds.



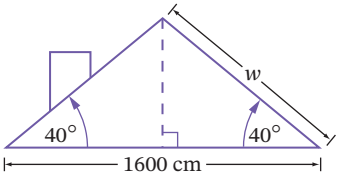
b The length of the ramp.



c The length of the support wire.



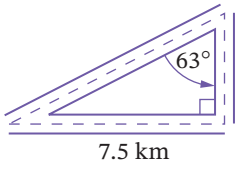
d The slant height of the roof.



e The length of:

i the shortest road

ii the longest road.



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EXAMPLE 15

8 A ladder rests against a wall. The foot of the ladder is 355 cm from the wall and makes an angle of 63° with the ground. How long (to the nearest cm) is the ladder? PS

9 A supporting wire is attached to the top of a flagpole. The wire meets the ground at an angle of 51° and the flagpole is 15 m high. How far, to the nearest 0.1 m, from the base of the flagpole is the wire anchored to the ground? PS

10 A glider is flying at an altitude (height) of 1.5 km. To land, it descends at an angle of 18° to the ground. How far, to the nearest 0.1 km, must the glider travel before landing? PS



Shutterstock.com/ Colombo Nicola

11 A shooter aims directly at a target, but just before firing, the rifle is lifted 1° off target. The shot misses the target by 67 mm. How far is the shooter standing from the target? Select **A**, **B**, **C** or **D**. PS

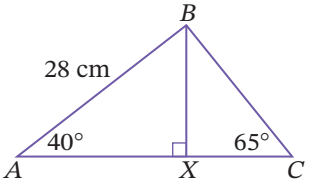
A 1169 mm **B** 3838 mm **C** 3839 mm **D** 6701 mm

12 A hot air balloon is anchored to the ground by a rope. When it drifts 20 m sideways, it makes an angle of 75° with the ground. How long is the rope (to one decimal place)? PS

13 For $\triangle ABC$, $\angle A = 40°$, $\angle C = 65°$, $AB = 28$ cm and BX is perpendicular to AC. Calculate, correct to one decimal place, the length of:

a BX

b CX



Investigation

Finding an angle, given a trigonometric ratio

You will need: a ruler, compasses and a calculator.

1 Copy and complete this table, calculating each ratio as a decimal, correct to 3 decimal places.

θ	sin θ	cos θ	tan θ
0°			
15°			
30°			
45°			
60°			
75°			
90°			undefined

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2 Can you work out why there is no answer for tan 90°?

3 What are the minimum and maximum values of sin θ?

4 What are the minimum and maximum values of cos θ?

5 Is there a pattern between the values of sin θ and cos θ?

6 Check the value of sin 30° by constructing a right-angled triangle with one angle that is 30°, measuring the opposite side and hypotenuse and dividing them.

7 Check the value of tan 45° by constructing a right-angled triangle with one angle that is 45°, measuring the opposite and adjacent sides and dividing them.

8 Use the table to estimate each trigonometric ratio and check your estimate using a calculator.

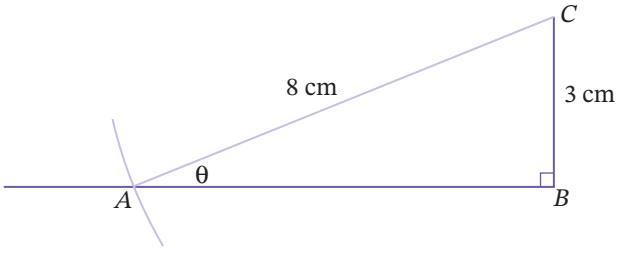
a sin 80° **b** cos 34° **c** tan 55°

9 If $\sin\theta = \frac{3}{8}$, find the value of the unknown angle θ, correct to the nearest degree:

a by using the table and estimating (change $\frac{3}{8}$ to a decimal first)

b using a calculator to guess-and-check

c constructing a right-angled triangle with one angle θ, opposite side 3 cm and hypotenuse 8 cm, then measuring the size of θ.



10 If $\cos\theta = \frac{2}{5}$, find the value of the unknown angle θ, correct to the nearest degree:

a by using the table and estimating (change $\frac{2}{5}$ to a decimal first)

b using a calculator to guess-and-check

c constructing a right-angled triangle with one angle θ, adjacent side 2 cm and hypotenuse 5 cm, then measuring the size of θ.

11 If $\tan\theta = \frac{7}{10}$, find the value of the unknown angle θ, correct to the nearest degree:

a by using the table and estimating (change $\frac{7}{10}$ to a decimal first)

b using a calculator to guess-and-check

c constructing a right-angled triangle with one angle θ, opposite side 7 cm and adjacent side 10 cm, then measuring the size of θ.

4.07 Finding an unknown angle

Worksheets
Finding an unknown angle
Trigonometry problems

Puzzles
Trigonometry squaresaw
Trigonometry: Finding angles
Trigonometry equations 2

Presentation
Applications of triangles

A scientific calculator can be used to evaluate a trigonometric ratio such as sin 38°, but it can also be used to find an unknown angle, θ, if the trigonometric ratio of the angle is known, for example, if sin θ = 0.9063.

An unknown angle can be found using the sin^{-1}, cos^{-1} and tan^{-1} keys on the calculator. These are called the **inverse sin**, **inverse cos** and **inverse tan** functions, found by pressing the SHIFT or 2nd F key before the sin, cos or tan key.

Example 16

a If sin θ = 0.9063, find angle θ, correct to the nearest degree.

b If tan X = 3.754, find angle X, correct to the nearest minute.

c If cos α = $\frac{4}{7}$, find angle α, correct to the nearest degree.

Solution

a sin θ = 0.9063

θ = 64.9989...°

≈ 65°

On a calculator: SHIFT sin 0.9063 =

b tan X = 3.754

X = 75.0837 ...°

= 75° 5′ 1.62″

≈ 75° 5′

SHIFT tan 3.754 =

° ′ ″ or D°M′S

c cos α = $\frac{4}{7}$

α = 55.1500...°

≈ 55°

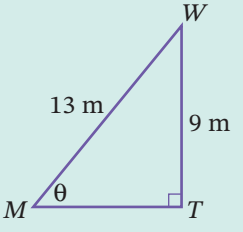
SHIFT cos 4 $\frac{\square}{\square}$ 7 =

Note: $\frac{\square}{\square}$ or $\frac{a}{b/c}$ is the fraction key.

Videos
Finding an unknown angle
Trigonometry

Example 17

Find the size of angle θ, correct to the nearest degree.



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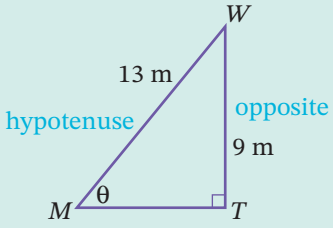
Solution

SOH, CAH or TOA?
The known sides for θ are the opposite (O) side and the hypotenuse (H), so use sin.

$\sin\theta = \frac{9}{13}$

$\theta = 43.8130...°$ SHIFT sin 9 ÷ 13 =
$\approx 44°$

From the diagram, an angle size of 44° looks reasonable.



Finding an unknown angle in a right-angled triangle

1 Identify the 2 known sides and decide whether to use the sin, cos or tan ratio.

2 Write an equation using the ratio, the angle and the 2 sides as a fraction.

3 Use the calculator's inverse trigonometric function to find the size of the angle.

Example 18

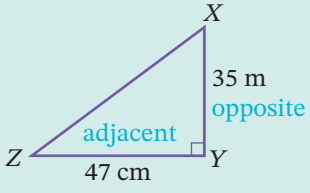
$\triangle XYZ$ is right-angled at Y, with $XY = 35$ cm and $YZ = 47$ cm. Find $\angle Z$, correct to the nearest degree.

Solution

Sketch a diagram.
SOH, CAH or TOA?
The known sides to $\angle Z$ are the opposite (O) and the adjacent (A), so use tan.

$\tan Z = \frac{35}{47}$

$Z = 36.6743...°$
$\approx 37°$
From the diagram, an angle size of 37° looks reasonable.



SHIFT tan 35 ÷ 47 =

EXERCISE 4.07 ANSWERS ON P. 619

Finding an unknown angle U F PS

EXAMPLE 16

1 Find the size of angle θ, correct to the nearest degree.

a $\cos\theta = 0.76$ **b** $\tan\theta = 2.0532$ **c** $\sin\theta = \frac{\sqrt{3}}{2}$

d $\tan\theta = 6$ **e** $\sin\theta = \frac{7}{8}$ **f** $\cos\theta = \frac{13}{15}$

g $\sin\theta = \frac{1}{10}$ **h** $\cos\theta = \frac{1}{\sqrt{2}}$ **i** $\tan\theta = \sqrt{3}$

j $\cos\theta = 0.1352$ **k** $\tan\theta = 8.836$ **l** $\sin\theta = \frac{1}{4}$

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2 Find the size of angle A, correct to the nearest minute.

a $\tan A = \frac{15}{7}$ **b** $\sin A = 0.815$ **c** $\cos A = \frac{4}{5}$

d $\cos A = 0.9387$ **e** $\tan A = \frac{19}{20}$ **f** $\cos A = \frac{3}{10}$

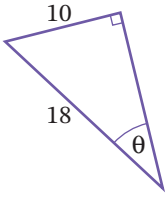
g $\sin A = \frac{5}{11}$ **h** $\sin A = 0.88$ **i** $\tan A = 15.07$

j $\cos A = \frac{1}{7}$ **k** $\tan A = \frac{\sqrt{2}}{2}$ **l** $\sin A = \frac{7}{9}$

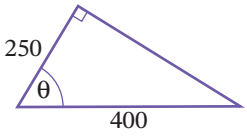
EXAMPLE 17

3 Find θ, correct to the nearest degree.

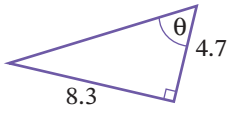
a



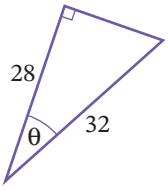
b



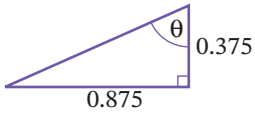
c



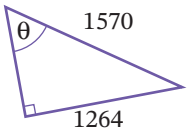
d



e



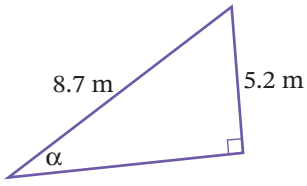
f



4 Find the size of angle α, correct to the nearest minute. Select the correct answer **A**, **B**, **C** or **D**.

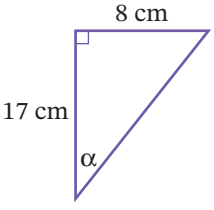
A 30° 52′ **B** 30° 53′

C 36° 42′ **D** 36° 43′

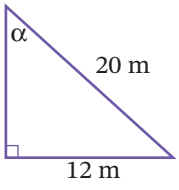


5 Find the size of angle α, correct to the nearest minute.

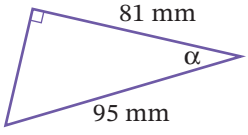
a



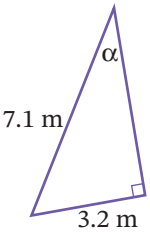
b



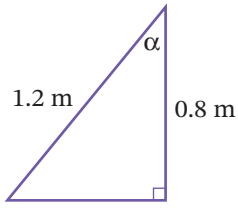
c



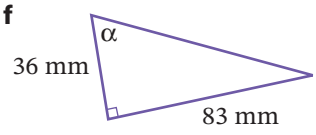
d



e



f

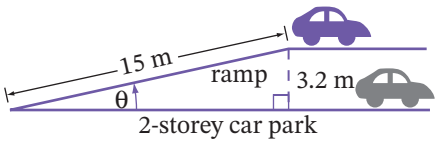


Foundation Standard Complex

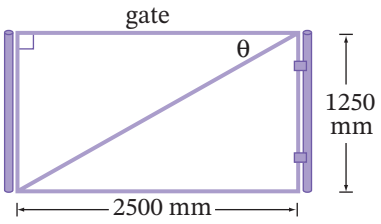
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6 Find the size of angle θ, correct to the nearest degree.

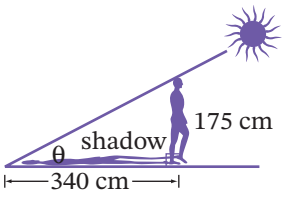
a



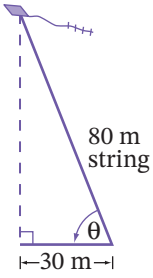
b



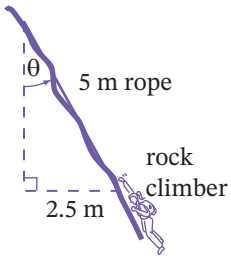
c



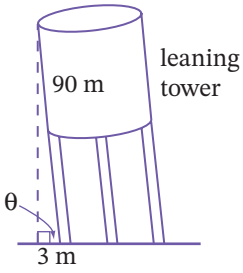
d



e



f



7 In $\triangle XYW$, $\angle X = 90°$, $XY = 8$ cm and $XW = 10$ cm. Find $\angle W$, correct to the nearest degree.

EXAMPLE 18

8 $\triangle HTM$ is right-angled at T, $HM = 45$ m and $MT = 35$ m. Find $\angle M$, correct to one decimal place.

9 In $\triangle FGH$, $\angle G = 90°$, $GH = 3.7$ m and $FH = 19.5$ m. Find the size of angle F, correct to the nearest minute.

10 In $\triangle CDE$, $\angle C = 90°$, $CD = 54$ mm and $CE = 42$ mm. Find $\angle E$, correct to the nearest minute.

For questions **11** to **17**, write answers correct to the nearest degree.

11 A stretch of freeway rises 55 m for every 300 m travelled along the road. Find the angle at which the road is inclined to the horizontal. PS

12 A ladder 20 m long is placed against a building. If the ladder reaches 16 m up the building, find the ladder's angle of inclination to the building. PS

13 An aircraft is descending in a straight line to an airport. At a height of 1270 m, it is 1500 m horizontally from the airport. Find its angle of descent to the horizontal. Select **A**, **B**, **C** or **D**. PS

A 32° **B** 40° **C** 50° **D** 58°

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14 A tree 8.5 m high casts a shadow 3 m long. What is the angle of the Sun from the ground? PS

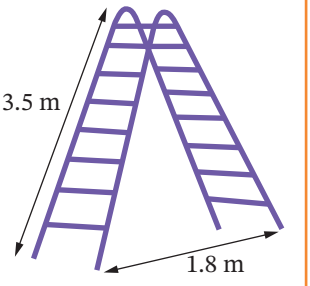
15 At a resort, an artificial beach slopes down at a steady angle. After walking 8.5 m down the slope from the water's edge, the water has a depth of 1.6 m. At what angle is the beach inclined to the horizontal? PS

16 A pile of wheat is in the shape of a cone that has a diameter of 35 m and measures 27 m up the slope to the apex. Calculate the angle of repose of the wheat (the angle the sloping side makes with the horizontal base). PS

17 A ship is anchored in water 40 m deep by a 65 m anchor chain. Find the angle at which the chain is inclined to the sea floor. PS

18 The arms of this fold out ladder are 3.5 m long. When fully spread, the feet of the ladder are 1.8 m apart.

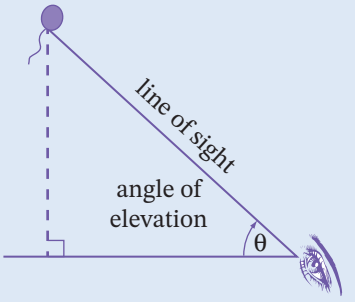
Calculate, correct to the nearest minute, the angle between the arms of the ladder when fully spread.



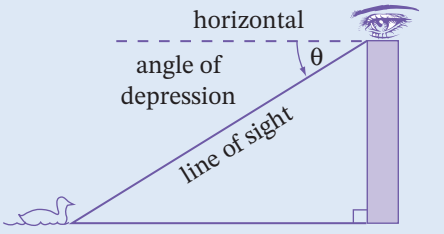
4.08 Angles of elevation and depression

Angles of elevation and depression

The **angle of elevation** is the angle of looking **up,** measured from the horizontal.



The **angle of depression** is the angle of looking **down,** measured from the horizontal.



Problems involving angles of elevation and depression usually require the tan ratio in their solutions.

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An instrument for measuring an angle of elevation or depression is a **clinometer**. It is like a protractor with a sighting tube attached.

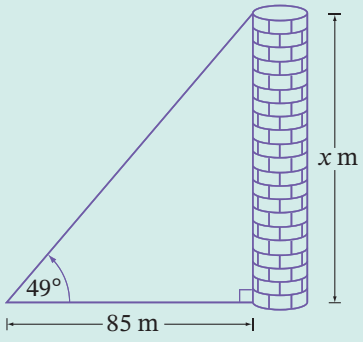


Getty Images/VW Pics

4.08

Example 19

The angle of elevation of a chimney is 49° at a distance of 85 m from its base. Find the height of the chimney, to the nearest metre.



Solution

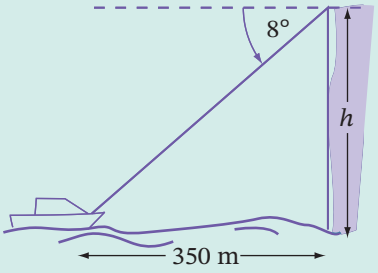
Let the height be x metres.

$$\tan 49° = \frac{x}{85}$$
$$x = 85 \tan 49°$$
$$= 97.7813 \ldots$$
$$\approx 98$$

The height of the chimney is 98 m.

Example 20

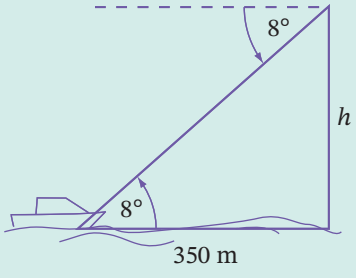
The angle of depression of a boat from the top of a cliff is 8°. If the boat is 350 m from the base of the cliff, calculate the height of the cliff, correct to the nearest metre.



Videos
Angles of elevation and depression
Angle of depression 1
Angle of depression 2

Solution

By alternate angles on parallel lines, the angle of elevation of the top of the cliff from the boat is also 8°.



$$\tan 8° = \frac{h}{350}$$
$$h = 350 \tan 8°$$
$$= 49.1892...$$
$$\approx 49$$

The height of the cliff is 49 m.

Alternative method:
The third angle in the triangle (adjacent to the angle of depression) = 90° – 8° = 82°.

$$\tan 82° = \frac{350}{h}$$

$$h = \frac{350}{\tan 82°}$$

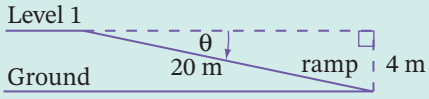
$$= 49.1892 \ldots$$

$$\approx 49$$

The height of the cliff is 49 m.

Example 21

The ramp from one level to the next in a car park is 20 m long and drops 4 m. Find the angle of depression of the ramp, to the nearest degree.



Solution

$$\sin\theta = \frac{4}{20}$$

$$\theta = 11.5369 \ldots°$$

$$\approx 12°$$

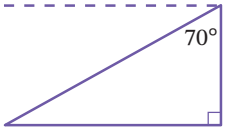
The angle of depression of the ramp is 12°.

EXERCISE 4.08 ANSWERS ON P. 619

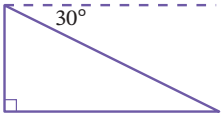
Angles of elevation and depression U F PS

1 Copy each diagram, mark the angle of elevation θ and find its size.

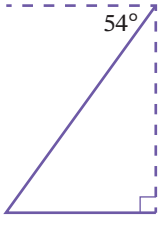
a



b

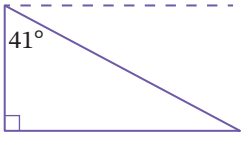


c

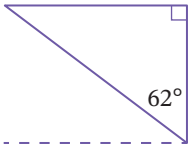


2 Copy each diagram, mark the angle of depression θ and find its size.

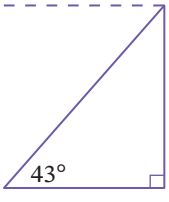
a



b



c

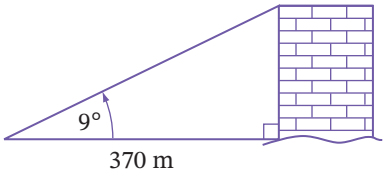


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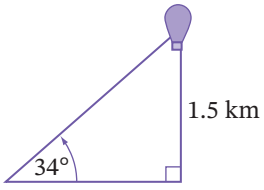
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3 Emily stands 370 m from the base of a building. Using a clinometer, she finds that the angle of elevation of the top is 9°. Find the height of the building, to the nearest metre.

EXAMPLE 19

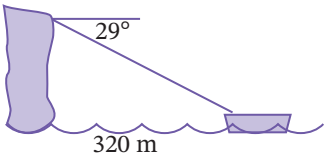


4 The angle of elevation of a weather balloon at a height of 1.5 km is 34°. How far (to the nearest metre) is the observer from being directly under the balloon?



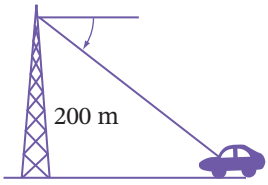
5 A raft is 320 m from the base of a cliff. The angle of depression of the raft from the top of the cliff is 29°. Find the height of the cliff, to the nearest metre. PS

EXAMPLE 20



6 From the top of a 200 m tower, the angle of depression of a car is 48°. How far is the car from the foot of the tower? PS

7 A 275 m radio mast is 1.7 km from a school. Find, correct to the nearest degree, the angle of elevation of the top of the mast from the school. PS



EXAMPLE 21

8 A monument 24 m high casts a shadow 20 m long. Calculate, correct to the nearest degree, the angle of elevation of the sun at this time of day. PS

9 In a concert hall, Ethan is sitting 20 m from the stage by line of sight. He is also 5 m above the level of the stage. At what angle of depression is the stage? Answer to the nearest minute. PS

10 A plane is 340 m directly above one end of a 1000 m runway. Find, correct to the nearest minute, the angle of depression to the far end of the runway. PS



Shutterstock.com/ verzellenberg

11 An observer 174 cm tall is standing 11.6 m from the base of a flagpole. The angle of elevation to the top of the flagpole is 43°. How high is the flagpole, to the nearest cm? PS

12 A flagpole is mounted on top of a tall building. At a distance of 250 m from the base of the building, the angles of elevation of the bottom and top of the flagpole are 38° and 40° respectively. Calculate the height of the flagpole, correct to one decimal place. PS

13 A news helicopter hovers at a height of 500 m. The angles of depression of a fire moving in the direction of the helicopter are first 10° and then 15°. How far (to the nearest metre) has the fire moved between the 2 observations? PS

14 The angle of elevation to the bottom of a transmission tower on a hill from an observer 1.8 km away from the base of the hill is 5°. The angle of elevation to the top of the tower from the observer is 6.8°. Find the height of the tower to the nearest metre. PS

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4.09 Bearings

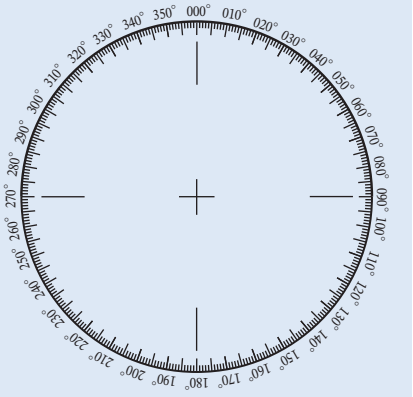
Bearings are used in navigation. A **bearing** is an angle measurement used to describe precisely the direction of one location from a given reference point.

Worksheets
A page of bearings
NSW map bearings
16 points of the compass

Puzzle
Bearings match-up

True bearings

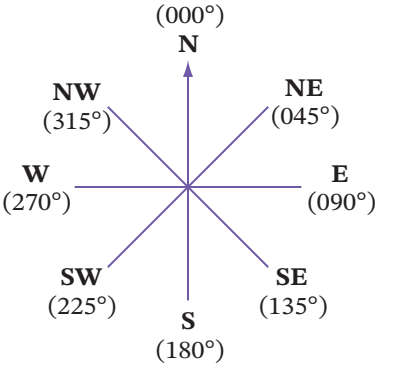
True bearings, also called **three-figure bearings**, use angles from 000° to 360° to show the amount of turning measured **clockwise from north 000°**. Note that the angles are always written with 3 digits.



True bearings

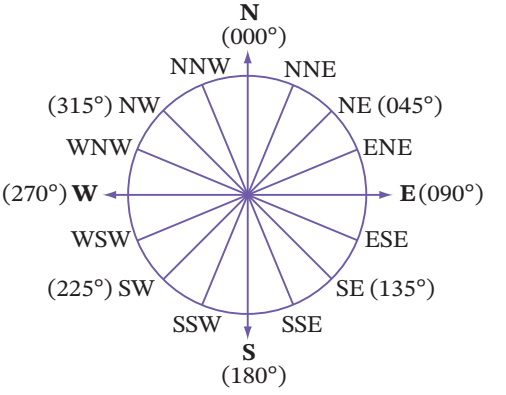
This **compass rose** shows the true bearings of 8 points on the compass. A bearing of due east is 090°, while a compass direction of southwest (SW) is 225°.

- Bearings from 000º to 090º are in the NE quadrant
- Bearings from 090º to 180º are in the SE quadrant
- Bearings from 180º to 270º are in the SW quadrant
- Bearings from 270º to 360º are in the NW quadrant



This diagram shows the 16 points of the compass.

N = north	NE = northeast
E = east	SE = southeast
S = south	SW = southwest
W = west	NW = northwest
NNE = north-northeast	ENE = east-northeast
ESE = east-southeast	SSE = south-southeast
SSW = south-southwest	WSW = west-southwest
WNW = west-northwest	NNW = north-northwest



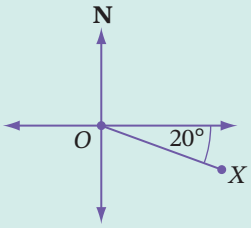
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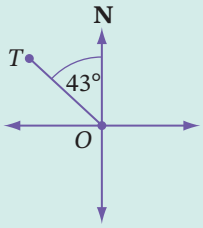
Example 22

Find the true bearing of each point from O.

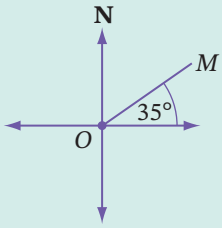
a



b



c



Solution

a Bearing of X from O is $90° + 20° = 110°$.

b Bearing of T from O is $360° - 43° = 317°$.

c Bearing of M from O is $90° - 35° = 055°$

This is sometimes written as 110°T to indicate that it is a True bearing.

Must be written as a 3-digit angle

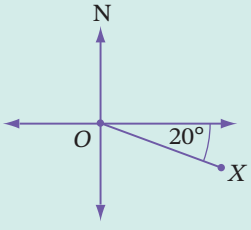
Compass bearings

Compass bearings use acute angles measured from the North-South axis towards East or West, such as N20°E or S57°W. They are named according to the quadrant in which they are located, and are used in land surveying.

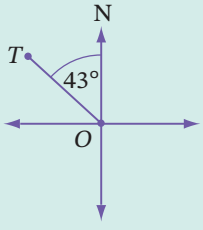
Example 23

Write the compass bearing of each point from O.

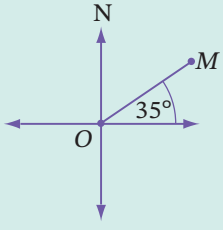
a



b



c



Solution

a X is in the southeast (SE) quadrant. From south, it is $90° - 20° = 70°$ towards east.

The compass bearing of X from O is S70°E.

b T is in the northwest (NW) quadrant. From north, it is 43° towards west.

The compass bearing of T from O is N43°W.

c M is in the northeast (NE) quadrant. From north, it is $90° - 35° = 55°$ towards east.

The compass bearing of M from O is N55°E.

Video
Three-figure bearings

4.09

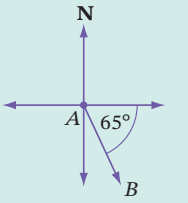
Video
Three-figure bearings

Example 24

Sketch point B on a compass rose if B has a bearing of 155° from A.

Solution

Draw the compass rose on the point where the bearing is measured from.
155° is between 90° and 180°, so B is in the southeast (SE) quadrant.
$155° - 90° = 65°$, so B is 65° from east (E).



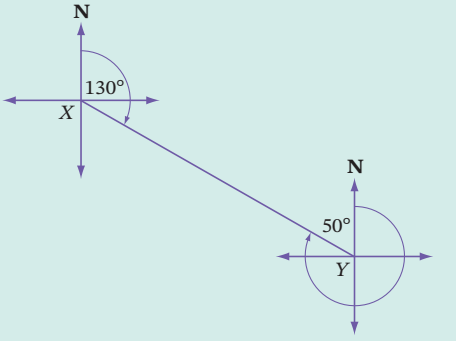
Video
Three-figure bearings

Example 25

The bearing of Y from X is 130°. What is the bearing of X from Y?

Solution

Sketch the bearing of Y from X.
On the same diagram, draw a compass rose at Y and find $\angle NYX$.
$\angle NYX = 50°$ (co-interior angles, $NX \parallel NY$)
$\therefore$ Bearing of X from $Y = 360° - 50°$
$= 310°$



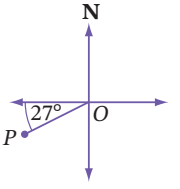
EXERCISE 4.09 ANSWERS ON P. 619

Bearings U F R C

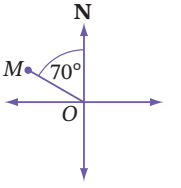
EXAMPLE 22

1 Find the true bearing of each point from O. C

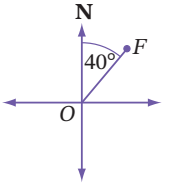
a



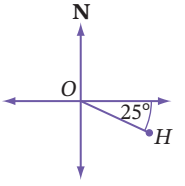
b



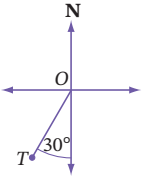
c



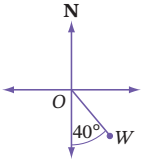
d



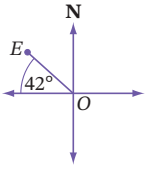
e



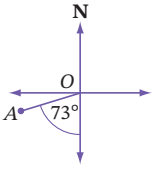
f



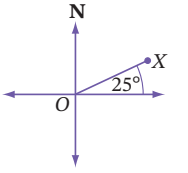
g



h



i



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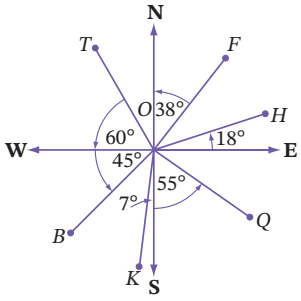
2 What is the bearing of each direction or point from O? C

a N **b** E **c** S

d W **e** F **f** Q

g T **h** B

i H **j** K



3 Find the compass bearing of each point from O in Question **1**. C

4 Find the true bearing of each compass bearing. C

a SW **b** E **c** NE **d** ESE

e NNW **f** W **g** SE **h** WSW

5 Sketch each bearing on a compass rose.

a 220° **b** 060° **c** 260° **d** 125°

e N10°W **f** S87°W **g** S9°E **h** N32°E

6 **a** What is the compass direction halfway between northwest and west?

b What is the compass bearing of this compass direction? C

7 Sketch P on a compass rose if P has a bearing of:

a 132° from T **b** 260° from M **c** 335° from X **d** 010° from K

8 If the bearing of P from A is 060°, what is the bearing of A from P? R

9 The bearing of T from Y is 100°. What is the bearing of Y from T? R

10 What is the (smallest) angle between:

a S and SW? **b** NE and SE? **c** E and NW?

d NE and SSW? **e** E and SSW? **f** SW and WNW?

11 The compass bearing of H from M is N65°W. Find the compass bearing of M from H. R C

12 Draw a diagram for each situation described. C

a A plane flies on a bearing of 280° for 150 km and then another 250 km on a bearing of 080°.

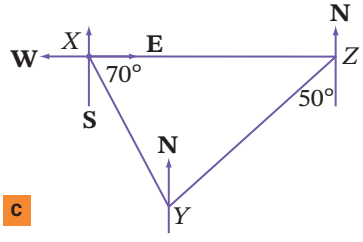
b A cyclist travels 15 km due east and then 20 km on a S45°W bearing.

13 For this diagram, find the true bearing of: C

a Y from Z

b X from Z

c Y from X



14 For the above diagram, find the compass bearing of: C

a X from Y

b Z from Y

c Z from X

4.09

EXAMPLE 23

EXAMPLE 24

EXAMPLE 25

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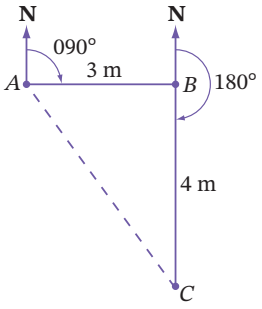
Investigation

Compass walks

You need: a directional compass and a tape measure or trundle wheel.

This activity can also be done in the classroom using scale drawings, graph paper, a ruler and a protractor.

A triangular walk



1. Starting at A, walk due east for 3 m to B.
2. From B, walk due south for 4 m to C.
3. How far is C from A?
4. What is the bearing of:
   - **a** A from C?
   - **b** C from A?

A square walk

1. Starting at P, walk at a bearing of 045° for 8 m to Q.
2. From Q, walk at a bearing of 315° for 8 m to R.
3. From R, walk at a bearing of 225° for 8 m to S.
4. How far is S from P?
5. What is the bearing of:
   - **a** P from S?
   - **b** S from P?

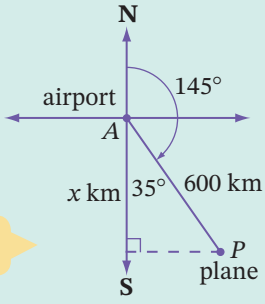
A pentagonal walk

1. Starting at U, walk at a bearing of S50°E for 4 m to V.
2. From V, walk at a bearing of N40°E for 7 m to W.
3. From W, walk at a bearing of N40°W for 4.8 m to X.
4. From X, walk at a bearing of west for 4.5 m to Y.
5. How far is Y from U?
6. What is the bearing of:
   - **a** U from Y?
   - **b** Y from U?

Problems involving bearings 4.10

Example 26

A plane leaves an airport and flies 600 km on a bearing of 145°. How far south of the airport is the plane (to the nearest km)?



Solution

To find x, create a right-angled triangle.

Let x km = distance south

$\therefore \angle SAP = 180° - 145°$ (angles on a straight line)

$= 35°$

$\cos 35° = \frac{x}{600}$

$x = 600 \cos 35°$

$= 491.4912 \ldots$

≈ 491

The plane is 491 km south of the airport.

Videos
Bearings
Trigonometry 2
True bearings

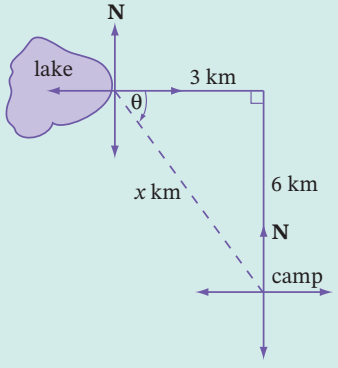
4.10

Example 27

From camp, Sumaiyaa walks due north for 6 km, then 3 km due west to a lake.

a How far is Sumaiyaa from the camp, correct to the nearest 0.1 km?

b What is the bearing of the camp from the lake (to the nearest minute)?



Solution

a Let x = distance from camp.

$x^2 = 6^2 + 3^2$ by Pythagoras' theorem

$= 45$

$x = \sqrt{45}$

$= 6.708...$

≈ 6.7 km

Sumaiyaa is 6.7 km from the camp.

b Note angle θ in the diagram.

$\tan\theta = \frac{6}{3}$

$\theta = 63.434...°$

$= 63° 26' 5.82"$

$\approx 63° 26'$

Bearing of camp from lake

$= 90° + 63° 26'$

$= 153° 26'$

Video
Problems involving bearings

EXERCISE 4.10 ANSWERS ON P. 620

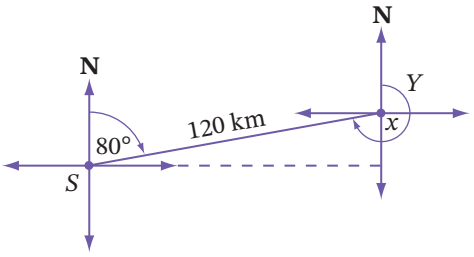
Problems involving bearings U F PS C

EXAMPLE 26

1 A yacht leaves Sydney and sails 120 km on a bearing of 080°.

a How far north (to the nearest km) of Sydney is the yacht?

b What is the bearing of Sydney from the yacht?

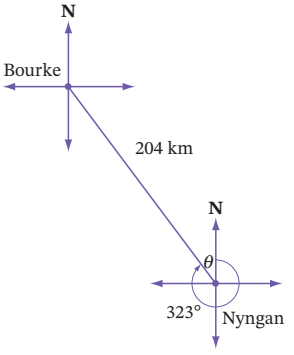


2 Declan leaves Nyngan and drives 204 km to Bourke. The bearing of Bourke from Nyngan is 323°.

a Find the value of θ.

b How far north (to the nearest km) of Nyngan is Bourke?

c What is the bearing of Nyngan from Bourke?



3 The distance 'as the crow flies' from Sydney to Wollongong is 69 km. If the bearing of Wollongong from Sydney is 205°, calculate:

a how far south Wollongong is from Sydney (correct to one decimal place)

b how far east Sydney is from Wollongong (correct to one decimal place)

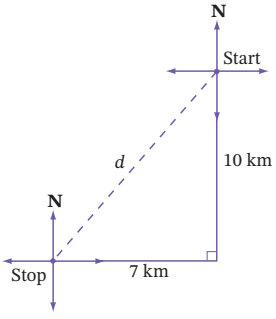
c the compass bearing of Sydney from Wollongong.

EXAMPLE 27

4 Jana cycles 10 km due south, then 7 km due west.

a How far (correct to one decimal place) is Jana from her starting point?

b What is her true bearing from the starting point, correct to the nearest degree?



5 A triathlete cycles 20 km on a SSE bearing to the finish line.

a How far (to the nearest km) has the triathlete travelled in a southerly direction?

b What is the compass bearing of the starting point from the finish line? PS C

6 A hiking group walks from Sandy Flats to Black Ridge (a distance of 20.9 km) in the direction 078°. They then turn and hike due south to Rivers End, then due west back to Sandy Flats. How far have they hiked altogether (to the nearest 0.1 km)? PS C

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7 A triangular orienteering run starts at Alpha and passes through the checkpoints of Bravo and Charlie before finishing at Alpha. Bravo is 8.5 km due east of Alpha, and Charlie is 10.5 km due south of Bravo. PS C

a Calculate, correct to 3 decimal places, the distance from Charlie to Alpha.

b Find the compass bearing of Alpha from Charlie to the nearest degree.

8 A plane takes off from Darwin at 10:15 a.m. and flies on a bearing of 150° at 700 km/h. PS C

a How far (to the nearest km) due south of Darwin is the plane at 1:45 p.m.?

b What is the bearing (correct to the nearest degree) of Darwin from the plane?

9 A fishing trawler sails 30 km from port on a bearing of S60°E until it reaches a submerged reef. How far (to the nearest km) is the port: PS C

a north of the reef?

b west of the reef?

10 2 racing pigeons are set free at the same time. The first bird flies on a course of 040° while the second bird flies on a course of 130°. PS C

a The first bird flies 200 km until it is due north of the second bird. Find their distance apart, correct to 2 decimal places.

b How far has the second bird flown?

11 2 horse riders start from the same stable. The rider of the black horse goes due west for 5.5 km and stops. The rider of the chestnut horse travels in a direction of N57°W until he is due north of the black horse. How far did the rider of the chestnut horse travel? Answer correct to 2 significant figures. PS C

12 2 ships leave from the same port. One ship travels on a bearing of 157° at 20 knots. The second ship travels on a bearing of 247° at 35 knots. (1 knot is a speed of 1 nautical mile per hour.) PS C

a How far apart are the ships after 8 hours, to the nearest nautical mile?

b Calculate the bearing of the second ship from the first, to the nearest minute.

13 A plane flies 1200 km on a NE compass bearing and then changes direction to a bearing of NW and flies for a distance of 450 km. PS C

a How far, correct to the nearest km, is the plane from its starting point?

b What is the bearing of the plane from its starting point, to the nearest minute?

14 A fishing boat leaves Sydney Harbour on a compass bearing of ESE and after 15 nautical miles, changes direction to a bearing of SSW and travels a further 10 nautical miles.

Find the bearing of Sydney from the boat's final position, to the nearest minute. PS C

Power plus ANSWERS ON P. 620

1 **a** Copy and complete each pair of trigonometric ratios, correct to 3 decimal places.

i $\sin 20° =$ _____, $\cos 70° =$ _____ **ii** $\sin 47° =$ _____, $\cos 43° =$ _____

iii $\sin 55° =$ _____, $\cos 35° =$ _____ **iv** $\sin 85° =$ _____, $\cos 5° =$ _____

b What do you notice about each pair of answers in part **a**?

c What do you notice about each pair of angles in part **a**?

d If $\cos 30° \approx 0.8660$ and $\sin \theta \approx 0.8660$, what is the value of θ?

e Copy and complete each equation.

i $\sin 75° = \cos$ ____ **ii** ____ $80° = \cos 10°$

iii $\cos$ ____ $= \sin 72°$ **iv** $\sin 30° =$ ____ $60°$

v $\cos 65° = \sin$ ____ **vi** $\sin$ ____ $= \cos 58°$

f Copy and complete this general rule: $\sin x = \cos$ (_____).

g Use a right-angled triangle with one angle x and sides a, b and c to prove that the above rule is true.

2 A plane is flying at an angle of 15° inclined to the horizontal.

a How far will the plane have to travel along its line of flight to increase its altitude (height) by 500 m?

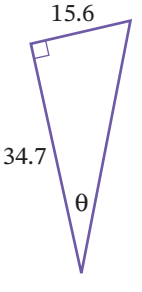
b At what angle must the plane climb to achieve an increase in altitude of 500 m in half the distance needed at an angle of 15°?

3 If $\sin 30° = \frac{1}{2}$, find, as a surd, the value of:

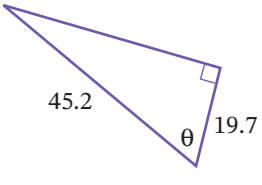
a $\cos 30°$ **b** $\tan 30°$

4 Find angle θ, correct to the nearest second.

a



b



5 By drawing an appropriate triangle, prove that:

a $\tan 45° = 1$ **b** $\sin 45° = \frac{1}{\sqrt{2}}$ **c** $\cos 45° = \frac{1}{\sqrt{2}}$

6 Sketch a right-angled triangle with side lengths a, b and c, where c is the hypotenuse. Let θ be one of the acute angles. Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

CHAPTER 4 REVIEW

Quiz
Language of maths 4

Puzzle
Trigonometry crossword

Language of maths

adjacent	angle of depression	angle of elevation	bearing
clinometer	compass bearing	cosine (cos)	degree (°)
denominator	horizontal	hypotenuse	inverse ($^{-1}$)
minute (′)	opposite	right-angled	second (″)
sine (sin)	tangent (tan)	theta (θ)	true bearing
trigonometry	trigonometric ratio	unknown	vertical

1 When measuring angle size, what is a **second** and what is its symbol?

2 What word means 'next to'?

3 Which side of a right-angled triangle is fixed and does not depend on the position of an angle?

4 What are the first 2 letters of the Greek alphabet?

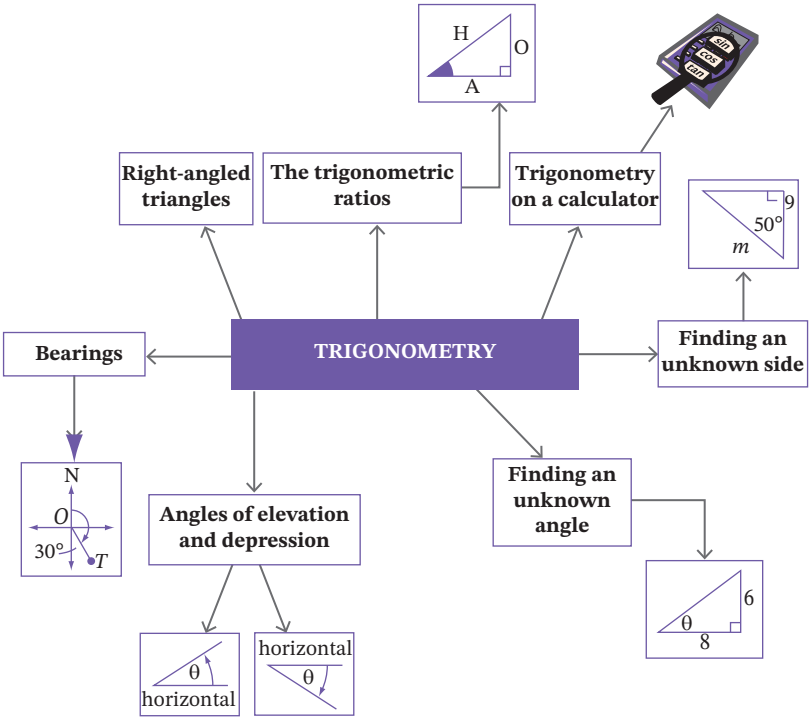
5 The word **minute** has an alternative pronunciation and meaning. What is its alternative meaning?

6 What does **inverse** mean and how is it used in trigonometry?

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Trigonometry (Advanced)



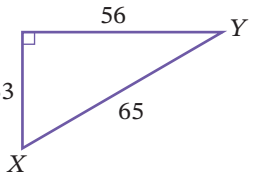
Quiz
Test yourself 4

TEST YOURSELF 4

ANSWERS ON P. 621

4.02

1 For this triangle, write as a fraction:



a $\sin Y$ **b** $\tan Y$

c $\sin X$ **d** $\cos X$

4.02

2 If $\sin \alpha = \frac{36}{85}$, find the values of $\cos \alpha$ and $\tan \alpha$ as fractions. (Draw a diagram.)

4.04

3 Round each angle to the nearest degree.

a 64° 27′ **b** 25° 43′ **c** 12° 8′ 50″

4.04

4 Round each angle to the nearest minute.

a 50° 19′ 26″ **b** 31° 55′ 55″ **c** 64° 18′ 30″

4.04

5 Evaluate each expression, correct to 4 decimal places.

a $\cos 32°$ **b** $\sin 50° 9'$ **c** $\tan 8° 45'$

d $200 \tan 18°$ **e** $14 \sin 87° 40'$ **f** $\frac{13}{\cos 18° 27'}$

4.04

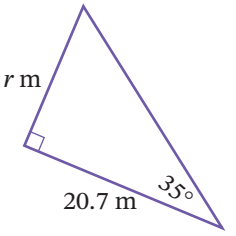
6 Convert each angle size to degrees and minutes, correct to the nearest minute.

a 45.8° **b** 33.175° **c** 5.346°

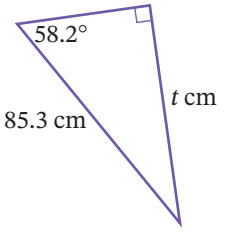
4.05

7 Find the value of each variable, correct to 2 decimal places.

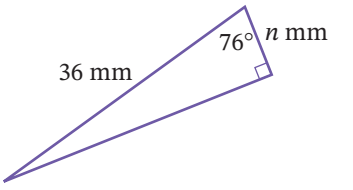
a



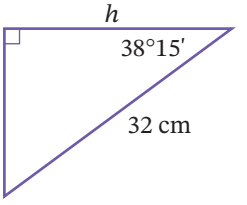
b



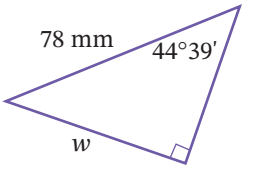
c



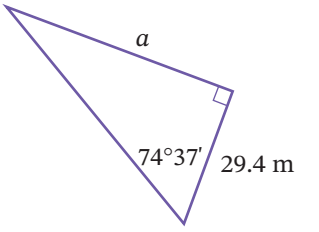
d



e



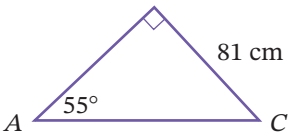
f



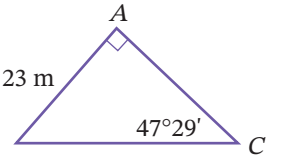
4.06

8 For each triangle, find the length of side AC, correct to one decimal place.

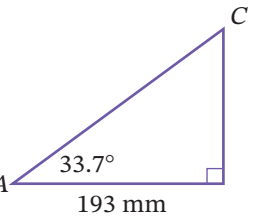
a



b



c



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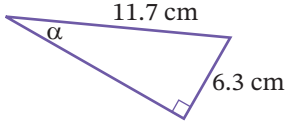
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9 Find θ, correct to the nearest minute. 4.07

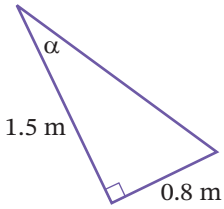
a $\tan\theta = 2.57$ **b** $\cos\theta = \frac{4}{7}$ **c** $\sin\theta = \frac{1.5}{1.6}$

10 Find the size of angle α, correct to the nearest degree. 4.07

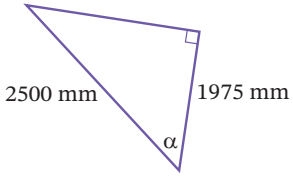
a



b



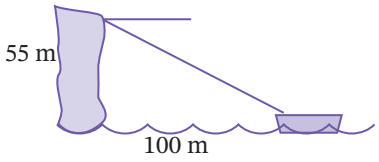
c



11 In $\triangle AEC$, $\angle C = 90°$, $CE = 3.9$ m and $AE = 4.2$ m. Find $\angle A$, correct to the nearest minute. 4.07

12 The angle of elevation of a tower roof is 26° at a point 400 m from its base. Find the height of the tower correct to the nearest metre. 4.08

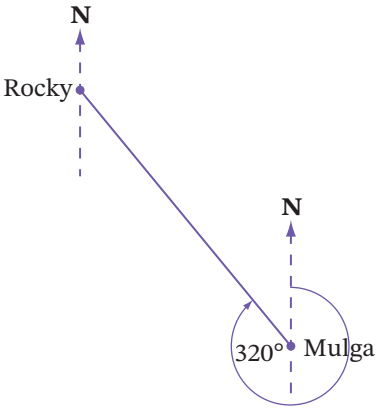
13 Find the angle of depression (correct to the nearest degree) of a boat that is 100 m from the base of a 55 m cliff. 4.08



14 What is the true bearing of: 4.09

a Rocky from Mulga?

b Mulga from Rocky?



15 Two planes leave an airport at the same time. The first travels on a bearing of N63°E at 500 km/h. The second travels on a bearing of S27°E at 400 km/h. 4.10

a How far apart are the planes after 2 hours, to the nearest km?

b Calculate, correct to the nearest degree, the compass bearing of the first plane from the second plane.

16 A boat sails from Perth on a compass bearing of SSW for 12 nautical miles and then changes direction to a bearing of WNW and sails a further 42 nautical miles. 4.10

a How far is the boat from Perth?

b What is the true bearing of the boat from Perth?

c On what bearing will it have to sail on its way directly back to Perth?

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